(4th, 5th, 9th) ⇒ How can we use it?

Page No.
Date 13/03/25

a) Total no. of possible partitions

Sort of recursive formula

(Bell no) $B_{n+1} = \sum_{k=1}^{n} S(n,k)$ ← Stirling 2nd No

$S(n,k) = \binom{n}{k} B_k$

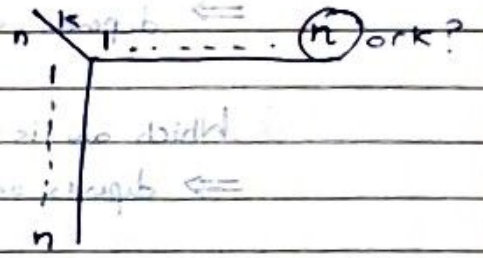
i) Non-overlapping
ii) Non-empty (atleast 1) } partitions

Assume $(n-1)$ data points, k partitions I have

$\Rightarrow k\,S((n-1),k) + S((n-1),(k-1)) = S(n,k)$

External Evaluation Method

b) All pairs sent to,



⇒ TP, TN, FP, FN

Rand Index = $\frac{TP + TN}{TP + TN + FP + FN}$ $({}^{n}C_2)$ [0, 1]

To Do List — Not To Do List ← Success in not to do list doesn't matter as it is very big

here TN is very large

⇒ TP to be removed (??) (X) ← here we consider TP

Jaccard Index = $\frac{|\text{Intersection}|}{|\text{Union}|}$ [0, 1] ← Here Intersection & Union is taken betn ideal & the new cluster formed ⇒ it talks of how similar

$= \frac{TP}{TP + FP + FN}$

FP ∈ mine, ∉ ideal
FN ∈ ideal, ∉ mine

predicts ⇒ it's bad on small positive cases (If FN↑)

[0,1]

Precision = $\frac{TP}{TP + FP}$ ← out of all positive predicted how many correct

for together pairs like in the cluster out of this how much correct

(i.e Precision ⇐ recall)

[0,1]

Recall = $\frac{TP}{TP + FN}$ ← out of all positive cases how many identified

out of all possible pairs that exists in Ideal

Perfect Score: If all pairs marked as positive in my cluster ⇐ We can't go either of them separately

Page No:
Date: / /

F1 measure (Harmonic Mean) $0-1$

$$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

note: it is by (0/0) assumed to be 0

If one goes down it goes down significantly

⇓

We have Fowlkes & Mallow Index (FM)

$$\sqrt{\text{Precision} \times \text{Recall}}$$

Which one is better? similar measure

⇒ depends on application

Which one is best clustering method?

⇒ depends on context

Internal Measures

- (No ground truth)
- We can't go for TP, TN, FP, FN
- go across all clusters, within each cluster, compute score

Method1 for each data point (S)

$$\square = \sum_{i=1}^{K} \frac{1}{|C_i|} \sum_{j \in C_i} S(x_j^i)$$

$S(x_j^i) \in [-1, 1]$

evaluation measure for that clustering

S:

- $A(x_j^i)$ Average intra cluster distance: $\sum_{\substack{m \in C_i \\ m \neq j}} d(x_j^i, x_m^i)$
- Average distance to nearest cluster: $B(x_j^i)$ $\min_{\substack{n=1,2,\dots k \\ n \neq i}} \left(\frac{1}{|C_n|} \sum_{y \in C_n} d(x_j^i, y) \right)$

Silhouette Coefficient

$\to (-1, 1)$

$$S = \frac{B(x_j') - A(x_j')}{\max(B(x_j'), A(x_j'))}$$

$B(x_j')\uparrow$ & $A(x_j')\downarrow$ for points very well placed inside cluster → (1)

for " not placed very well → (-1) → $B(x_j')\downarrow$, $A(x_j')\uparrow$

What's the problem with this?

→ all pair wise dist req. after clusters formed (computation high)

→

Method 2: Go for each cluster → (This one tells us how tight & well separated the clusters are)

	C_1	C_2	...	C_k
C_1				
C_2				
⋮				
C_k				

	Diameter
C_1	
C_2	
⋮	
C_k	

- how to define distance bet^n 2 clusters?

(1) → any choice

→ for points method specific

(2) we calculate the diameter as well

↳ It tell us spread out of the cluster {maximum distance within cluster}

(??) To determine bad cluster

(A) • Pick up lowest inter cluster distance / Highest Diameter ← called dunn Index

$$= \frac{\text{min inter-cluster}}{\text{max intra cluster}}$$

C_i C_j ← look at centroid something

Avg Intra cluster distance : $A(C_j)$ (here first, go for any)

$$\frac{A(C_i) + A(C_j)}{d(C_i, C_j)}$$ ← for computing distance bet^n 2 clusters & then do (A) again

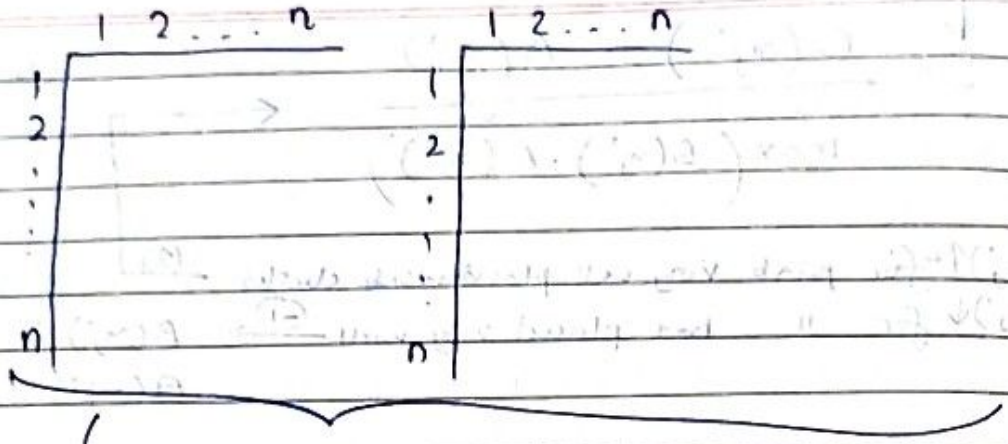
Q. Should we go with 1 clustering method? (No)

↳ Can I club together multiple clustering algo? (Yes)

Page No:
Date

Combining Multiple Clustering Algorithm → Clustering Ensemble

Each A_i gives diff clusters (or may be same)



- What can we infer from this for each pair?

Method 1 ⇒ Average for the final matrix
: Similarity measure
Distance = 1/Similarity

Method 2 ⇒ Graph clustering algo
→ here we can go for graph
i) Build graph
ii) With distances as the weight
iii) Since now we have got the whole graph, we can apply any clustering algo (ex remove edges etc) to get our desired no. of clusters

- Go through all m matrices → by taking avg
- Compute the final Similarity matrix
- then use any clustering algo

a)

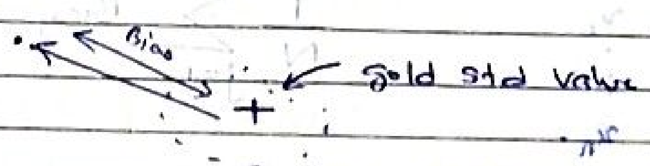
b)

Single data

a) Quiz 2 : April 2nd / 3rd , Project : April 2nd Week (Friday)
Mid Sem : Small Weight
End Sem : Open Notes

\# Bias Variance Trade Off

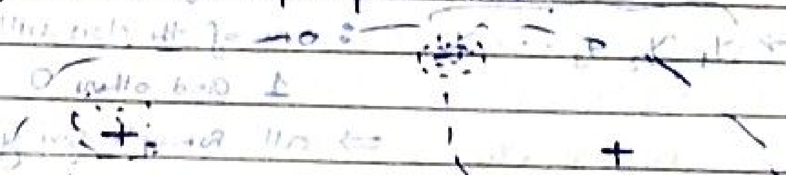
b) • Making predictions in 2D - space
• Bias : how far away from gold truth



→ how unsteady our predictions is

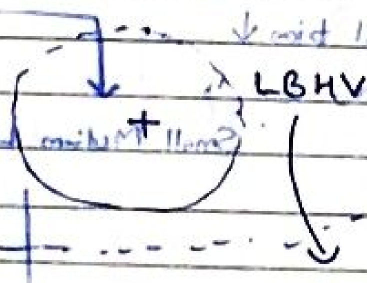
Variance : measures how far all points (to each other?)

Q What is multiple predictions?



low bias
low variance
classifier } ← IDEAL (Difficult to do)

high bias
low variance
classifier } ← complex patterns in data can't be understood {Underfitting}



• Overfitting to data hence not understand the complex patterns
• like on depending on small variations

Gold std value

Regression Case 1

c) x → y : (Build a model for x to predict y)

Single data point

Statistical bias

$\hat{y}_1 : M_1$ $\quad$ $Bias^2 = E\left[(y - E(\hat{y}))^2\right]$

$\hat{y}_2 : M_2$ (Since single point accounted hence not req) ← expectation

↑ expectation over all models

⋮

Q How to build the models (multiple)?

$\hat{y}_M : M_m$ ⇒ • random initialisation
• Sample data set (sampling with replacement)

Page No:
Date: / /

$$E(\hat{y}) = \frac{1}{M}\sum_{j=1}^{M} \hat{y}_j$$ ← Numerical attribute (0 – 100)

Case 2 : Multiple data points

x_1, x_2, ⋮, x_n

$$\frac{1}{N}\sum_{j=1}^{N} Bias_j^2$$ ⇐ What does it represent? $Bias^2$ for dataset (Test)

Classification: Single Data point

$x \rightarrow y_1\ y_2\ y_3 \cdots y_k$ → k classes : one of the class will be 1 and others 0

→ k classes indicate what? → each indicate probabilities (x belongs to each k classes partially) like fraction

⇒ for each classes considered ⇒ all sum upto give y

Total Bias ← What does this represent?

$$\Rightarrow \sum_{C=1}^{K}\left(P(y=C|x) - E(\hat{P}(y=C|x))\right)^2$$ ← ~~Numerical~~ Categorical (Red, Blue, Green)

What about for Ordinal?

Small Medium Large

⇒ don't go for one sol^n

⇒ Represent it in terms (i) of numeric / (ii) classification ~~Category~~

⇒ ~~Otherwise~~, represent diff bet^n Large - Med & Large - Small as same (hence ignore the ordinality)

⇒ if all bias ↓, then total bias ↓

else if bias one only we have, total bias not ↓

M: ĝ

assume equal spacing b/w categories, which might not be true

• Bias types: ~~Data Bias~~ considered

→ Statistical bias

dataset given is truly to be learnt

Fairness bias in Machine Learning

(iii) Ordinal metric: eg Cohen's Kappa with quadratic weighing

penalises larger errors more heavily

→ A low variance suggests that the models generally agree on the closer

Variance : (no picture of ideal point)

$\hat{y}$: prediction made (output prediction made)

$$\Rightarrow E\left[\left(\hat{y} - E(\hat{y})\right)^2\right] = \frac{1}{n / n-1} \sum_{i=1}^{n} (y_i - E(\hat{y}))^2$$

diff data points only? or y_i for m models?

E(ŷ): across all models. n: population variance; $n-1$: for sampled variance

across all ~~data points~~ all models : as in all possible $\hat{y}_i$ (?)

(Test)

For a model, we look into bias + variance + irreducible error

$$y = f(x) + \epsilon$$

$f(x)$: true value has some error associated with it

ϵ: error added (built into it) : (built in noise in measurement) → (not systemic error) → mean = 0, variance = σ^2

(Single data point across all multiple models)

$E\left[(\hat{y} - y)^2\right] \Rightarrow E\left[\left(\hat{y} - f(x) - \epsilon\right)^2\right]$

$E[(\hat{y} - y)^2]$: expectation of squared error (expected error); $\hat{y}$: prediction made; ϵ: (irreducible error); in the expansion, $\hat{y} - f(x) = a$ and $\epsilon = b$

What were you trying to do here?

$$\rightarrow E\left[(\hat{y} - f(x))^2\right] + E\left[\epsilon^2\right] - 2E\left[(\hat{y} - f(x))\epsilon\right]$$

$T_1 = E[(\hat{y} - f(x))^2]$: Model Error; $T_2 = E[\epsilon^2] = \sigma^2$ (variance of noise term); $T_3 = 2E[(\hat{y} - f(x))\epsilon]$

$T_3 \rightarrow 2E(\hat{y} - f(x))\,E(\epsilon)$, with $E(\epsilon)$ = mean : 0, so $T_3 = 0$. Why independent?

$$(\hat{y} - f(x))^2 = (\hat{y} - E(\hat{y}) + E(\hat{y}) - f(x))^2 \quad \text{(A)}$$

where $a = \hat{y} - E(\hat{y})$ and $b = E(\hat{y}) - f(x)$.

Taking expectation on (A):

$$= E\left[(\hat{y} - E(\hat{y}))^2\right] + E\left[(E(\hat{y}) - f(x))^2\right] + 2ab$$

- a^2 (variance): diff data points? or diff models?
- b^2 ((bias)^2) ← E(bias^2) = bias^2 (Since it is one data point)
- $2ab$: need it to disappear

$$(2ab) : E\left[2(\hat{y} - E(\hat{y}))(E(\hat{y}) - f(x))\right]$$

- $\hat{y} - E(\hat{y})$ → 0 (This is mean value)
- $E(\hat{y}) - f(x)$ → c: As expectation over all models remain constant & same with f(x) i.e. constant

Page No:
Date: 19/03/25

Type of classification pbs

a) Hierarchical classification — labels are not list (not discussed) ex. — more trickier to evaluate

Person — Person

Family

Relation

Parents

Father Mother

(not discussed)

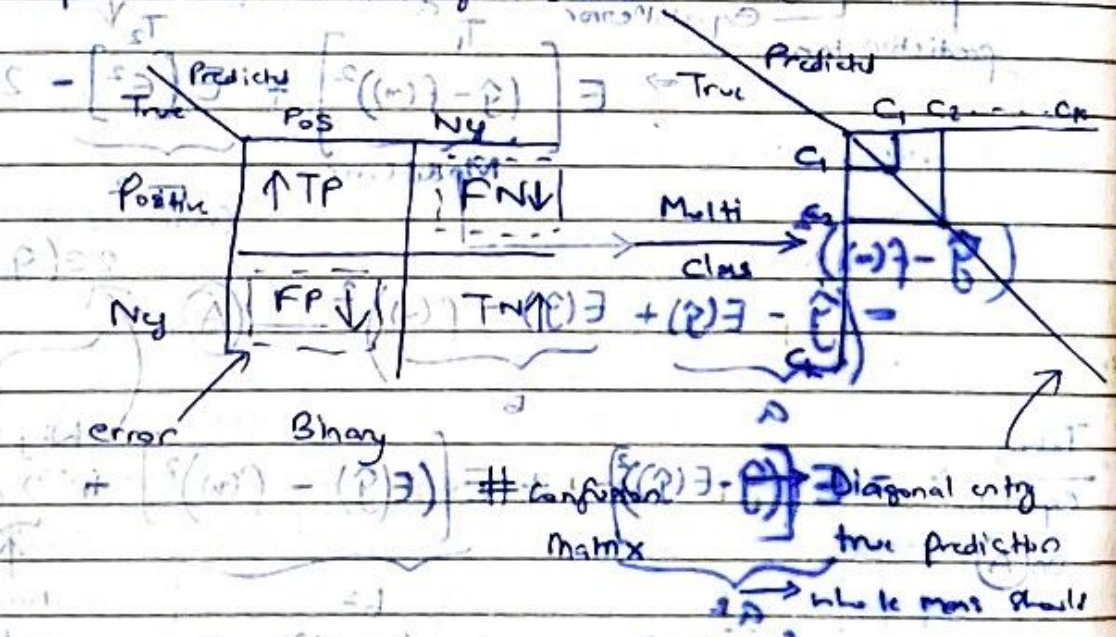
(d) Fuzzy classification — membership across all labels (Sort of probability) distribution

b) Binary — out of the class

Multiclass ← Discussed

Multilabel — More than 1 class

c) After a model is trained, given a dataset, it will predict outputs based on either of 4 classes:



d) Depends on the classification problem, (ensure) concentrate ground either low FN or medium (FN & FP) diagonals is okay

e) example. Medical Test

FP : person don't have disease, but predicted to have

FN : person has disease, but predicted not to have

Which one should be low: FP or (FN)?

→ depends on context like type of disease, medical treatment

example. Spam Email

FP : Not Spam email but went to Spam

FN : actually Spam !! but predicted not spam

Which one should be low? (FP) or FN?

example. What classification problem trying to solve?

⇒ Test Batter (play long innings) } ← is the ball risky?

T20 Batter (score runs quickly)

↑ can I hit the ball? & get runs?

(Is it swarsh?)

TP	: TP
TN	: TN
FP	: FP
(FN) : risky ball; but predicted not risky	scoring but pred : (FN) — Ct did not scoring

How to measure classification ~ Similar

f) if model accuracy (≈) it doesn't imply performance is same as FN is high

(measures how often the model's predictions are correct overall)

→ How can we get it high? (if FN is 99% make all FN) (Smtg)

→ Accuracy = $\frac{TP + TN}{\text{All predictions}}$

To get accuracy high → we can make predictions when we are only confident to get TP ↑

⇒ One single measure doesn't give Overall idea of the model

Example. Model confidence ignored (high pb ⇒ the class)

C1 C2 C3

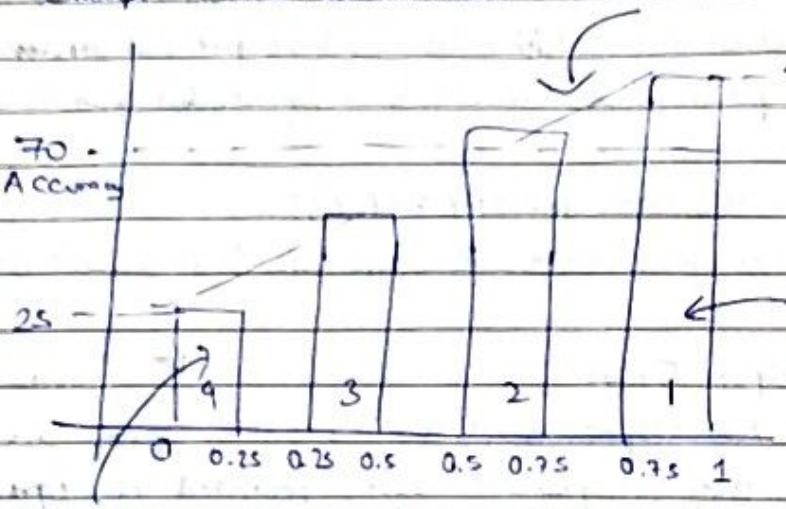
P1 (P2) P3 — # Deterministic Decoding

M1 → 0.90 }
M2 → 0.75 } Are these 2 same? even if they report C2 (No)

Page No:
Date: / /

example: Model Calibration

Well Calibrated model

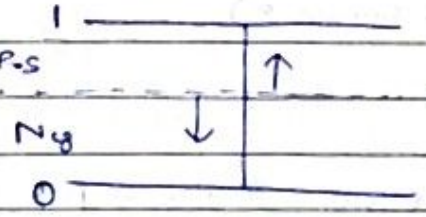


problems which I know & attempted (confident)

problems which I didn't know but attempted (not confident)

- Higher confidence ⇒ higher chance of success

example Threshold Tuning



→ have no idea of how accuracy will change if we go above the threshold

Or if you taken

- what is threshold have for?

To classify objects

- Problems related to Precision & Recall

(most are negative)

→ high Precision } — what does it not capture?

→ high Recall } like we can make very rare positive cases but still have high precision

everything is positive

(CHECK!)

Page No:
Date: / /

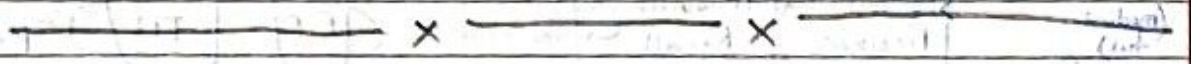
Specificity = $\frac{TN}{TN+FP}$ ← Since if mark all negative & FN not considered, it gives high value

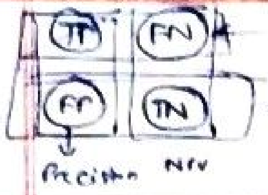
↑
- don't miss any negative class
- here ignoring false negative
- (So if mark all : negative)

Negative Predictive Value : $\frac{TN}{TN+FN}$ ← predict only when confident (negative part)

Make predictions only when confident
- very small no. of FN.
- then this value is high
- since predicts negative values mostly

⇒ if we mark them positive as well it won't matter



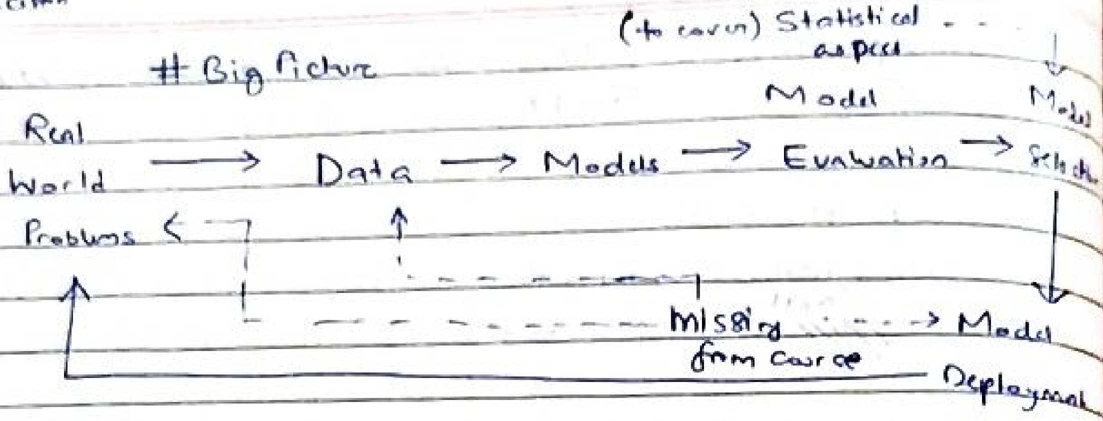


Evaluation Measure

FPR = 1 − Specificity
FNR = 1 − recall

Page No:
Date: 20/08/25

Big Picture



- How to prepare data (skipped) — data cleaning, data prep
- No domain knowledge — what's real world pb

few false cases → Reliability

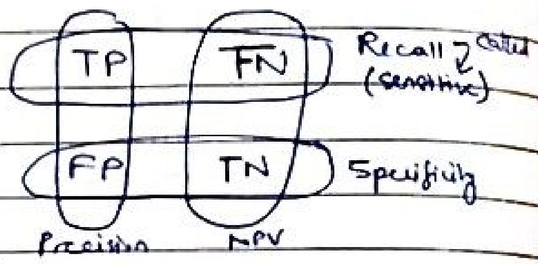
none of the positive things are missing → Coverage

Class \ Aspect of Prediction	Reliability	Coverage
Positive	(conservative action) Precision (cares FP)	(preventive action) Recall
Negative	NPV Aggressive (cares FN) Practice	~~Defensive practice~~ Specificity

(medical test) → Precision

Recall ↳ like covid → (I am rewarded for all TP)

Specificity → I am rewarded for all TN



aggressive practice
→ very much reliability for negative class
→ I might miss many neg
→ when I say neg, it is neg
→ example. Mass Recruitment (??)
T20 batter

→ I should not miss any neg
→ ex. judicial system (no innocent should be punished)

↱ Type 1 error : affects precision

(i) FPR (False positive Rate) = 1 − Specificity = $\frac{FP}{FP+TN}$

(ii) FNR (False negative Rate) = 1 − Sensitivity (Recall) = $FN / TP+FN$

↳ Type 2 error : affects recall

✱ The table talked about what measure I will take if I want to check for a classifier model with reliability or coverage for negative or positive predictions (confusion)

reliability : how much can you trust the model's prediction?

coverage : how well does the model capture all relevant cases?

good for focusing on positive class performance

→ helps evaluating a model performance, especially when classes are imbalanced (i.e. when positive cases are rare compared to negative)

req high recall, high precision

Precision Recall Curve — Closer to top right the better model

- Score for each point — [0,1]

Focused: How to decide threshold i.e. 0.8 → 1 ?
or 0.6 → 1 ?

- for each threshold, we have precision & recall ← this tells for only that threshold (not as in whole classifier performance)
  (so for that vary the threshold to get general idea)

- Sort the data points based on Score

(in this case R = 0) This is ideal classifier
(— perfect ranking it gives (nothing to deal with threshold)

Top Score — TP ↓

P_5

P_2

P_1

TN

Move the threshold a little down

(as it changes a little P=1)

P_{90}

P_7

Random peak value = % of positive instances

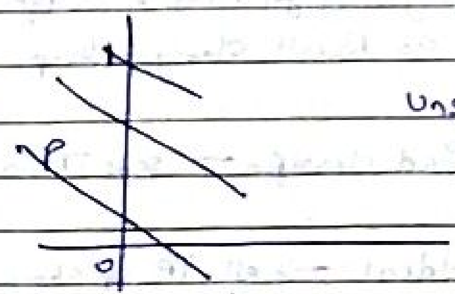
Random Classifier

→ Order is truly reflecting original ranking

→ In some interval → ← 90 → ← 10 →
Failure True

(initially unlucky/lucky & thereafter reflecting ideal classifier)
Unstable ⇐

2



Here we talk about area under the curve

Ideal = 1 ← (AUC)

(a good PR curve has greater AUC)

(NOTE: classifier that have a higher AUC on the ROC curve will always have a higher AUC on the PR curve as well)

Page No.
Date: / /

for good classifier – initial graph same as ideal cluster

for bad classifier – It is unstable initially extremely

Enam Sweet Point
- → Evaluation at low recall : all four same
- → When low recall : ideal & good same
- → high recall : All four diff
- → Complete high recall : only ideal has (1) position & (random good bad) together
- → extreme : all four same

Enam Sweet Spot : The point where adding more models doesn't significantly improve scoring but increases computation

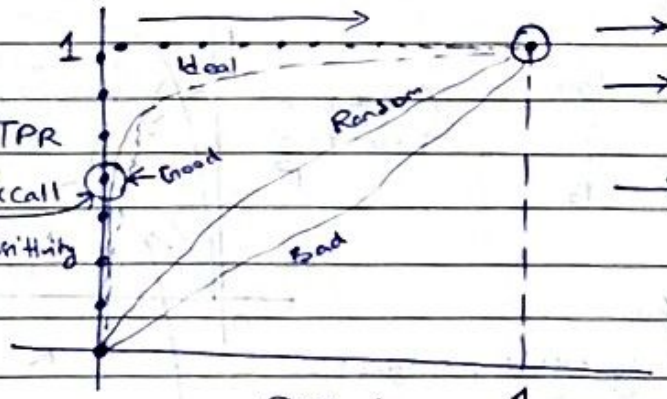
Receiver Operator Characteristic Curve — used for balanced dataset

↳ ideal point : top left (high TPR, low FPR)

→ no labels assigned yet
→ only scores assigned
→ So do something with threshold to measure if good classifier or not

good for overall class separation ability

Steps
→ initially threshold high
→ lower threshold gradually
→ mark all points gradually
→ how will ideal behave?



(Pointing point early)

... : ideal curve gradually increasing (here)

(but previously it is shooting up)

⇒ No jittery for random classifier as Recall slowly builds up

⇒ Bad classifier — says TP are at bottom

⇒ Ideal → all TP up above threshold & TN below

⇒ How good classifier picks up FP very quickly compared to prev curve (Precision Recall curve)

(for each possible threshold, there is a corresponding Recall & precision) → What is threshold in PRC?
→ Is it same point? (No) (Yes)
→ is it like for this threshold, PRC is this?

i) Increasing the threshold might increase Precision (fewer false positives) but it could lower Recall (more false negative)

ii) Goal in PRC is to find a threshold that either minimises FP or FN (depending on context)

(iii) ROC → depict tradeoff between specificity & sensitivity

(iv) Positive prediction accuracy ≡ Precision
Positive " completeness ≡ Recall

—— x —— x ——

Topics Covered Are:

a) K-Means & its variations
   (i) Spherical K-Means — how is centroid updated? cluster assigned?
   (ii) Constrained K-Means ↳ it's just the distance
   (iii) K-Means ++ — how is the score assigned? ↳ also preference given to points that are far from current centers
   (iv) With D^2 inequality
   (v) Privacy Preserving K-Means
   (vi) Global K-Means ↳ what does each can actually represent?
   (vii) Different cases of K-Means
   (viii) Alkin's Methods ← how upper, lower calculated?
   (ix) Anti-clustering-K-Means
   ↳ (check the proof property from notes)

Random → Global → K-Mean++

• Check for which path we get log k term

b) properties of clustering
   (i) Scale Invariance
   (ii) Richness property
   (iii) Consistency

clustering algorithms
   (i) k-clustering
   (ii) r-distance
   (iii) p^* distance

c) Clustering Measurement
   • External Evaluation Method — Rand, Jacard, F1
   • Internal " "

d) Bias Variance Trade Off

$(a, b) = ((a_1, a_2, \ldots a_n), (b_1, b_2, \ldots b_n))$

Page No:
Date: / /

Task To Do!

(e) Types of classification pb
Evaluation method for each class

(i) Read Note
(ii) Read the Extra Materials
(iii) Revise

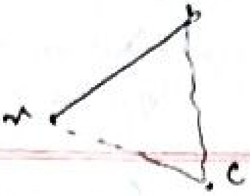
Extra Points Covered

a) WSSE: $\sum_{j=1}^{k} \sum_{x \in C_j} (x_i - y_j)^2$

b) K-Means ensures that cluster gets better at each step
⇓
How? – As in step 2, at every iteration we try to assign to a cluster to which,
$\min_j (x - c_j) \Rightarrow$ New SSE < Old SSE
also in step 3, when we update the cluster after assignment, the update ensures that the new centroid/representative is moved towards mean
⇒ within cluster distance for each point from its centroid reduces.

c) Spherical K-Means: ~~$x_i - c_j$~~ $\hat{x}_i$ ← unit vector
$x_i \in$ cluster c_j
(max)
We try to max $c_j \cdot \hat{x}_i$ i.e for that i for which, it will be maximum

d) Seed selection in K-Means++ via: $\text{Prob}(p_i) = \frac{\text{Score}(p_i)}{\sum_{d \in D} \text{Score}(p_d)}$
(Score = distance squared)
after first seed selection
max prob one taken as next centroid



e) Elkan's Method

5 matrices: upper bound } ← is it computed for current
lower bound } ↗ assignment?
membership — What was the eqn telling before?
distance to each centroid
Smallest distance centroid to current centroid

If $d(c_j) > 2d(v_i)$ —Yes→ ⓧ ← pick the min distance to current centroid from Task 5
↓No
$d(c_j - c_k) > 2d(v_i)$ —Yes→ ⓧ ← for each centroid (c, ≠ k) with current centroid
↓No
$d((v_i - c_j)) > 2d(v_i)$ —Yes→ ⓧ ← for each point — centroid &
↓No
Update

* Why random initialisation better than probabilistic initialisation?

⇒ Random Initialisation
- there is a possibility that they accidently land close to the optimal centroids
- limiting SSE to 2x the optimal

⇒ K Means++ initialisation
- despite being probabilistic, K-means++ prioritises points that are far away, reducing the risk of poorly placed centroids
- but in the worst case expected scenario, the SSE can go upto 8x the optimal
- This bound accounts for the fact that K-means++ has a probabilistic guarantee of good initialisation but doesn't guarantee a perfect centroids in all cases

Implications of the Probabilistic Update

The design of the probabilistic update ensure that:
- centres are well spreaded, reducing the chance of poor clustering
- The algorithm is less sensitive to outliers compared to purely random initialisation
- There is a theoretical guarantee on the quality of the initialisation relative to the optimal clustering → ($O(\log k)$ bound)

(Use when both FP & FN are important) → Why harmonic mean? → (it balances both metrics equally strongly that both need to be high for a good F_1 Score)

Date 21/03/25

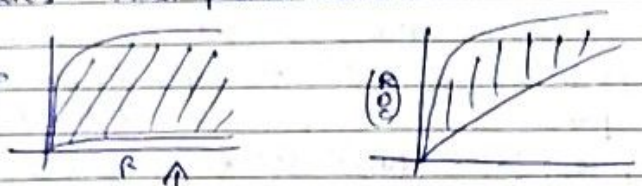
a) Another measure: F1 score (Combination of reliability & coverage)

↓

in terms of Recall & precision

→ random classifier – not good classifier

Case 1: When positive cases rate



$\left(2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right)$

F1 Score – most positive cases with high confidence

(both are equally important cases) → Sensitive to both P & R

→ F-β Score : general form

$(1+\beta^2) \times \frac{\text{Precision} \times \text{Recall}}{(\beta^2 \times \text{Precision}) + \text{Recall}}$

- weights given to reliability & coverage
- if $\beta < 1$: F-β = Precision : Accuracy of the positive predictions
  if $\beta > 1$: F-β = Recall

example (check once)

→ Pearson Correlation (Spearman)

Invariance Prop (")

b) Confusion Matrix for Multi Class Classification

True value → Actual class

Predicted value → Predicted class

each cell tells how many samples of C_i belongs to C_j (i.e how often a specific actual class was predicted as another class)

#Row: actual class

#Col: predicted "

False Negative

True label \ Predicted label	C_1	C_2	C_3	C_4
C_1	✓	×	×	×
C_2	×	✓		
C_3	×		✓	
C_4	×			✓

False Positive

What are the data points?

each C_{ij} tells no. of instances of actual class i predicted as class j

TN: Sum of all elements not in row i or column j

What will happen? # data points actually (belong to) in C_i (ideal)

What does this mean?
(i) if row adds sum → # predicted to be in C_i
(ii) if col adds sum
(iii) if row + col give total no. of data points

$\left(\sum f_i \neq N\right)$ ← (as can be counted in multiple class) (for multi class)

NOTE: for multiclass classification, we generally compute the measure per class (Σ measure) → then take avg

i) Accuracy (same

ii) Precision : (i) M

P_1 =

P_2 =

P_3

P_4 =

(ii) Weighted (importance of class proportional to its size) W

iii) Similarly for

iv) & Sen

v)

NOTE: for multiclass classification, which is the right measure?
we generally compute the measure per class (I think) → depends on application
↳ then take average as Precision, Recall & F1 fundamentally binary metrics

Page No:
Date:

i) Accuracy (same defⁿ)

ii) Precision : (i) Macro (each class equally imp)

$$P_1 = \frac{TP_1}{TP_1 + FP_1}$$

$$P_2 = \frac{TP_2}{TP_2 + FP_2}$$

each class equally important

$$\text{Macro} = \frac{1}{k}\sum_{i=1}^{k} P_i$$

k classes

$$P_3 =$$

$$P_4 =$$

- which is better?
→ weighted or macro
→ both have equal imp

(ii) Weighted $= \sum_{i=1}^{k} w_i P_i$

(importance of class proportional to its size)

$$w_i = \frac{r_i}{N} = \frac{Row_i}{N}$$

row corresponding to that particular class to total

iii) Similarly for recall → macro, weighted

often a specific actual class predicted as another class) iv) & same as F1-Score

{calculate F1 Score for per class}
(check once)
Then aggregate it : (i) micro avg (ii) macro avg (iii) weighted avg

v) Micro average — don't stick with any class

$$\boxed{\sum_{i=1}^{k} \frac{TP}{TP+FP}} \Rightarrow \frac{\sum_{i=1}^{k} TP_i}{\sum_{i=1}^{k}(TP_i + FP_i)}$$

data points actually (belong to) in C_1 (ideal)

= # predicted to be in C_1

$$\sum(\text{columns}) = \frac{\text{Correct prediction}}{N}$$

= Accuracy

↳ counted in multiple class)
* ↔ (for multi label)

→ each instance can have multiple correct labels
→ Y_i — set of true labels for instance i
→ $\hat{Y}_i$ — set of predicted labels for instance i
→ N — total no. of instances

Page No:
Date: / /

c) Multi Label Classification — (Metrics calculated per instance & then averaged over the dataset)

(i) Exact Match Ratio (∝ Accuracy)

No partial credit

$$\frac{1}{N} \sum_{i=1}^{N} I(Y_i == \hat{Y}_i)$$

↑ ideal set of labels

• 1 if exact match
↑ (I) gives

(ii) Hamming loss → measures the proportion of the instance-label pairs that are misclassified

→ doesn't work well if k is large (high reward for true negatives)

(If all 0, hamming loss high)

$$\frac{1}{N \times K} \sum_{i=1}^{N} \sum_{j=1}^{k} I(y_i^j \neq \hat{y}_i^j)$$

if belongs to K labels

⇒ No of labels high ⇒ High Hamming loss

(iii) Jaccard Index — more balanced than Hamming loss

(if Union is 1, Jaccard index low)

$$\frac{1}{N} \sum_{i=1}^{N} J_i \quad \text{where } J_i = \frac{|y_i \cap \hat{y}_i|}{|y_i \cup \hat{y}_i|}$$

(iv) Precision — We have to go point label (previously done on class level)
→ Precision for datapoint, for classification (→ binary), for class (→ for multi-class)

different from previous precision

$$\sum_{i=1}^{N} P_i \quad \text{where } P_i = \frac{|y_i \cap \hat{y}_i|}{|\hat{y}_i|}$$

NOTE: data point based is fourth way to define precision just like macro, weighted in multi-class

• If confusion matrix given for multi label, is it possible to know how many data points in C_1 class? (→ how to fill up? (difficult))
⇒ Ravi doesn't tell

Page No:
Date: / /

(v) Recall — Similarly done on point label

d) Pearson Correlation between X & Y } ← for binary
Matthews " Coefficient } → Multiclass also
↳ ≈ (Pearson but on computation of predicted & actual) (Smtg) Exists but not discussed

Ex. $A_i \leftarrow 0$ -ve class
1 +ve "
$P_i \leftarrow 1$ Predicted correctly

and for thru) # Some derivation
(a) $A_i^2 \rightarrow A_i$ (as squaring $A_i^2 == A_i$)
(b) $A_{i1} \rightarrow A_{mean}$
↑ all actual positive
$P_{i1} \leftarrow$ All predicted positive

↳ i.e true instances for class i
Note: $S_i \leftarrow$ Support for class i

Multiclass Classification Metrics
1) Multiclass Confusion Matrix (Precision, Recall, F1)
→ Averaging Strategies for Binary Metrics
(i) Macro average $= \frac{1}{C} \sum_{i=1}^{C} M_i$ Where M is the metric
↑ each class equally important

(ii) Micro average : Aggregate the contributions of all classes then compute avg globally
weight each ↑ instance equally

$$\text{Micro Precision} = \frac{\sum_{i=1}^{C} TP_i}{\sum_{i=1}^{C} (TP_i + FP_i)}$$

$$\text{Micro Recall} = \frac{\sum_{i=1}^{C} TP_i}{\sum_{i=1}^{C} (TP_i + FN_i)}$$

$$\text{Micro F1} = \frac{2 \times \text{Micro Precision} \times \text{Micro Recall}}{\text{Micro Precision} + \text{Micro Recall}}$$

Abhishek Anand (Sir's)

Page No
Date 24/03/25

→ Works with discrete feature : as key assumption is conditional feature independence given the class label

Bayesian Decision Theory

- for classification problem : supervised learning setting
- in here:
  - (i) probabilistic model
  - (ii) look for decision rule that would give optimal bayes error (lower bound on error)

↓

which gives notion of min error or min risk

example

Students: S1 S2 S3 . . . . Sn : n samples in my data

C1/C2 (C1 circled), C1/C2 (C2 circled), ... C1/C2 (C2 circled) : available information ⇓ hence, supervised learning

(pass or not pass)

each sample points belongs to one of C1 or C2 class

- Probability of students passing the course : we know ~~try to use bayes for~~
- $P(C_1)$ & $P(C_2)$: known — (it's fixed) — (A)
  - $P(C_1)$: prior prob
  - $P(C_2)$: prior prob
- Choose whether to take the course or not — (B)

(A), (B): Known

What should be the decision rule?

⇒ Choose $C_1 : P(C_1) > P(C_2)$

$C_2 :$ ~~$P(C_2) > P(C_1)$~~ otherwise

based on this ↓

- What's the error we are gonna ~~get~~ make? : i.e we choose C_1 but it was ideal to take C_2

Error prob : $P(error/s)$

$= \min(P(C_1), P(C_2))$ ← Why? ← (Think on it)

ex. $C_1 - P_1 = 0.7$

$C_2 - P_2 = 0.3$

error = 0.3

$[x, y]$

Page No:
Date: / /

feature independent class label

given a sample point, any passing prob not good
→ So we have features like background understanding

$S_i \vdash x_i$ ← feature

$P(x|c)$ ← Class conditional probability density

$p(c|s) = p(c|x)$ → we want to know, $P(C_1|x)$ or $P(C_2|x)$

What is the class a particular sample is gonna take?

This is the updated probability of the class given the observed features. It is what we are trying to predict/infer in a classification task

Classification Decision (The classifier selects the class C_k with the highest probability as the predicted class for the given set of features) ← (part of update rule)

$P(C_j, x) = P(C_j|x)\, p(x) = P(x|C_j) \cdot P(C_j)$

Bayes Formula → $P(C_j|x) = \frac{P(x|C_j)\, P(C_j)}{P(x)}$

current scenario ($P(x|C_j)$), past scenario ($P(C_j)$)

This is known posterior probability
⇒ How much prob. of this class changes provided x

Where $P(x|C_j)$ is 'likelihood of C_j wrt x'

probability of observing the given evidence (features) given that the class is true

⇒ This means the class C_j for which $P(x|C_j)$ is large is more likely to be true

Q What is $p(x)$? This is pb of obtaining the given evidence (features) ~~prior probability of~~

$P(C_j)$? prior probability of C_j: This is the pb of a class before considering any features

$p(x) = \sum P(x|C_j) \cdot P(C_j)$

- This is known as evidence
- This is generally not considered (the denominator part) as it is a normalising factor

NOTE: likelihood is how well the parameters of the model describe the outcome. The model with the optimal parameters maximize the probability.

(The probability is how probable an event or observation is for the given parameters of the model used)

Page No:
Date: / /

↓ sort of reassignment rule

∘ What would be my decision rule here?

$$\begin{cases} C_1 & \text{if } P(C_1|x) > P(C_2|x) \\ C_2 & \text{otherwise} \end{cases}$$

SAME →

$$\arg\max_{C_j} P(C_j|x) = \arg\max \frac{P(x|C_j)\,P(C_j)}{P(x)}$$

→ is it a single value? pb distribution? → yes, it gives the idea of class

we don't consider this ($P(x)$) — WHY? (Think)

Q What is my error probability?

$$P(\text{error}|x) = \begin{cases} P(C_1|x) & \text{if } C_2 \text{ decided class} \\ P(C_2|x) & \text{if } C_1 \text{ decided class} \end{cases} = \min\left(P(C_1|x), P(C_2|x)\right)$$

$$P(\text{error}) = \int_{-\infty}^{\infty} P(\text{error}, x)\, dx = \int_{-\infty}^{\infty} P(\text{error}|x)\, p(x)\, dx$$

(This is known as bayes error)

No decision rule will give me probability less than this ⇓ Resulting in total error to be min

→ Training sample independent to each other

* → Decision rule based on prior prob or posterior prob: The error associated with it gives us min value

• Assumption:

Generally th...

• S
• C
• Acti...

• Loss

Loss which has occurred when a decision has taken (α_i) & true class is ...

Avg over loss while taking decision (for a sample)

$$R = \int R(\alpha(x)|x)\, p(x)\, dx$$

↑ Overall risk

Page No:
Date 26/03/25

• Assumption: Population densities known

Generalize the idea

• $S \vdash x \in \mathbb{R}^d$ ← set of features

• $C = \{c_1, c_2, \ldots, c_k\}$: set of classes

• Actions / Decisions

$\alpha = \{\alpha_1, \ldots \alpha_m\}$ ← decision made

↳ • first it can be classes i.e. $\alpha_1 = c_1$

Type of decision $\alpha_2 = c_2$

↳ • there is no decision made i.e. no class chosen

• loss Fn: ← (previously we minimized the error. So this is related with it)

↳ cost associated with each action

loss which has occurred when a decision has taken (α_i) provided true class is c_j ⇒ $\lambda(\alpha_i \mid c_j)$ → $\lambda(\alpha_i|c_j)\, P(c_i|x)$

Note: Conditional loss

expectation

(This calculates the loss associated with a class provided (is true))

• Conditional Risk : defined over loss

avg over loss while taking decision (for a sample) ⇒ $R(\alpha_i \mid x) := \sum_{j=1}^{k} \lambda(\alpha_i|c_j) \cdot P(c_j|x)$

↑ expected loss associated with taking action α_i

here we want to minimize Conditional risk

$R = \int R(\alpha(x)|x) P(x)\, dx$ ⇒ we want to take α_i with reduced

Overall risk

$\alpha^* = \text{argmin}\,(R(\alpha_i \mid x))$

α_i

Page No:
Date: / /

• Specific Example.

$C = \{C_1, C_2\}$

$\alpha = \{\alpha_1, \alpha_2\} = \{C_1, C_2\}$

$\lambda_{ij} :- \lambda(\alpha_i \mid C_j)$ ← action, ← actual class

Conditional Risk

(i) $R(\alpha_1 \mid x) = \lambda_{11} P(C_1 \mid x) + \lambda_{12} P(C_2 \mid x)$

(action is α_1 provided C_1 is true; action is α_1 provided C_2 is true)

(ii) $R(\alpha_2 \mid x) = \lambda_{21} P(C_1 \mid x) + \lambda_{22} P(C_2 \mid x)$

{for given x my decision is α_1 which is C_1}

What's my decision?

$\Rightarrow \alpha_1 = C_1$: if $R(\alpha_1 \mid x) < R(\alpha_2 \mid x)$

$\hookrightarrow \lambda_{11} P(C_1 \mid x) + \lambda_{12}(P(C_2 \mid x)) < \lambda_{21} P(C_1 \mid x) + \lambda_{22} P(C_2 \mid x)$

Decision α_1

$\Rightarrow (\lambda_{21} - \lambda_{11}) P(C_1 \mid x) > (\lambda_{12} - \lambda_{22}) P(C_2 \mid x)$

posterior probabilities is weighted by loss/cost

What can we say about $\lambda_{21} - \lambda_{11}$ or $\lambda_{12} - \lambda_{22}$?

$\Rightarrow$ Are they positive or negative?

$\Rightarrow \lambda_{21}$ → cost occurred when decision is C_2 & true label is C_1

λ_{11} → loss when decision is C_1 & true label is C_1

$\lambda_{21} > \lambda_{11}$

$\Rightarrow$ Both terms are positive (> 0)

$\Rightarrow$ (we just make the decision based on posterior probability, no need of conditional Risk)

Construct another classifier (provided previous one)

→ It calculates the below ratio:

Decision2

$$\frac{P(x|c_1)}{P(x|c_2)} > \left(\frac{\lambda_{12} - \lambda_{22}}{\lambda_{21} - \lambda_{11}}\right) \frac{P(c_2)}{P(c_1)}$$

(for risk based error) ⇒ likelihood ratio; Threshold

c_2 is true

⇒ The option is: we can build another classifier based on likelihood ratio & say the sample point belongs to a class if likelihood ratio > (some threshold)

Q Assignment1: Derive D2 ~~[illegible]~~ & D1 (something)

Q Assignment2:

$$\lambda(\alpha_i|c_j) = \begin{cases} 0 & i=j \quad \text{Assign 0 loss to a correct decision} \\ 1 & i \neq j \quad \text{Assign unit loss to an incorrect decision} \end{cases}$$

Zero-one-loss function

Issue: All errors are equally costly } ← diff kind of incorrect decision happening (something)

→ (for eg. large and small if x is small, and we predicted large then also error is 1 & if it was predicted med then also which is not good as med prediction should have smaller diff error)

Summary

(i) In probabilistic model (. . . . . . .)

(ii) 2 different decision rule

Conditional risk becomes:

(i) $R(\alpha_i|\bar{x}) = \sum_{j=1}^{c} \lambda(\alpha_i|w_j) * P(w_j|\bar{x})$

$\Rightarrow R(\alpha_i|\bar{x}) = \sum_{j \neq i}^{c} P(w_j|\bar{x}) = 1 - P(w_i|\bar{x})$

(The very decision rule of Bayes' classifier automatically minimizes the conditional risk associated with an action.)

Page No:
Date: 28/03/25

- Bayesian Classifier ← What is like cont ? discrete here?
  - $S_i \vdash x_i \in \mathbb{R}^d$: cont feature vector (Known N)
  - $p(x|c_j)$, $P(c_j)$

min-error $\longrightarrow$ $C^* = \arg\max_{c_j} P(c_j|x)$

min-risk $\longrightarrow$ $\alpha^* = \arg\min_{\alpha_i} R(\alpha_i(x)|x)$

Binary Classification : $(\lambda_{21} - \lambda_{11})P(c_1|x) > (\lambda_{12} - \lambda_{22})P(c_2|x)$

Choose c_1 if it's true (min risk)

The another form of this is:

$$\frac{p(x|c_1)}{p(x|c_2)} > \left(\frac{\lambda_{12} - \lambda_{22}}{\lambda_{21} - \lambda_{11}}\right)\frac{P(c_1)}{P(c_2)}$$

(likelihood ratio)

Another Classifier (How obj is the classifier will be discriminant fn) Known as Discriminant Functions

Discriminant Functions as classifiers

a) We are defining discriminant function $g_i(x)$ for each class c_i

b) Q :- The aim is to define appropriate discriminant function (How is to be found)

⇓

for this, the decision rule is defined:

$$C^* = \arg\max g_i(x)$$

To minimize conditional risk

Choice 1 $\longrightarrow$ $g_i(x) = -R(\alpha_i|x)$

Choice 2 $\longrightarrow$ $g_i(x) = P(c_j|x)$ ← min error rate

} Few choices for discriminant function

Page No:
Date: / /

Choice 3 → Any monotonically increasing function of $g_i(x)$ in (1) & (2)

Thus can also be used

$$g_i(x) = \frac{p(x|c_i)\,p(c_i)}{p(x)}$$

$$g_i(x) = p(x|c_i)\,p(c_i)$$

$$g_i(x) = \ln p(x|c_i) + \ln p(c_i)$$

(since logarithm is monotonically increasing function, it doesn't change the result)

° for binary classifier, $\boxed{g(x) = g_1(x) - g_2(x)}$

$g(x) > 0 \longrightarrow c_1$

$g(x) < 0 \longrightarrow c_2$

c) if we have single / univariate normal density

$$p(x) = \frac{1}{(\sqrt{2\pi})\sigma} \exp\left(\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

$$p(x) \sim N(\mu, \sigma^2)$$

$$x \sim N(\mu, \sigma^2)$$

Mean: $\mu = E(x) = \int_{-\infty}^{\infty} x\, p(x)\, dx$

Variance: $\sigma^2 = E((x-\mu)^2) = \int_{-\infty}^{\infty} (x-\mu)^2\, p(x)\, dx$

d) Our interest is to look at multivariate case

Multivariate density (because x contains d dimensional vector)

Multivariate normal density →

$$p(\underline{x}) = \frac{1}{(2\pi)^{d/2}\,|\Sigma|^{1/2}} \exp\left[\frac{-1}{2}(\underline{x}-\underline{\mu})^{T}\,\Sigma^{-1}\,(\underline{x}-\underline{\mu})\right]$$

($1\times d$, $d\times d$, $d\times 1$)

$\begin{bmatrix} a & 0 \\ 0 & \beta \end{bmatrix}$

σ_{ii}: variance, σ_{ij}: covariance

$\sigma_{ij} = 0 \rightarrow x_i$ & x_j are statistically independent

$\sigma_{ij} = E\left((x_i - \mu_i)(x_j - \mu_j)\right)$

$x \in \mathbb{R}^d$

$\int x\, p(x)\, dx = \mu \in \mathbb{R}^d$

$\int (x-\mu)(x-\mu)^t p(x)\, dx = \Sigma \in \mathbb{R}^{d \times d}$ ← why cov should be $d \times d$ matrix?

→ positive, semidefinite & symmetric

$|\Sigma|$ ← determinant of covariance matrix

Σ^{-1} ← inverse of cov. matrix

What kind of classifier we will get?

given: $S_i \mid x_i \in \mathbb{R}^d$

$p(x|c_i)$, $p(c_i)$ (Here we are assuming class conditional density follows a multivariate normal density function)

- if we assume $p(\underline{x}|c_i) \sim N(\mu_i, \Sigma_i)$

Note: μ_i ← mean vector of class c_i

Σ_i ← covariance matrix of class c_i

for ith class, the values/(?) are coming from the Normal class having μ_i & Σ_i

(It defines a region in space s.t. sample points go there)

- let's say discriminant functions,

$$g_i(x) = \ln(p(x|c_i)) + \ln(p(c_i))$$

$$\Rightarrow g_i(x) = -\frac{1}{2}(x-\mu_i)^t \Sigma_i^{-1} (x - \mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln p(c_i) \quad \text{— (A)}$$

(We have covariance matrix as diagonal matrix)

Case I: $\Sigma_i = \sigma^2 I$ ← identity matrix of appropriate dimension

- What does it tell the relationship among features?
- Can anything be told about it?

Remarks

R1: (I) Σ_i is a diagonal matrix

(II) features are statistically independent,

i.e. $\sigma_{ij} = 0$ when $i \neq j$

↑ cov. both features x_i & x_j

...ally independent

(iii) all features have the same variance σ^2

R2: $|\Sigma_i| = (\sigma^2)^d$ ← Σ_i same for all class in LDA but in QDA (it's not)

R3: $\Sigma_i^{-1} = \left(\frac{1}{\sigma^2}\right) I$ (like $-\frac{d}{2}\ln 2\pi$ & $-\frac{1}{2}\ln|\Sigma_i|$)

(Whatever terms not depended on i is not considered)

$g_i(x) \neq -\frac{1}{2}(x-\mu_i)^t$ ~~as it is~~ hence to reduce computation ← same for all class

With the particular $\Sigma_i = \sigma^2 I$, we want to look at exp (A),

$$g_i(x) = \frac{-1}{2\sigma^2}(x-\mu_i)^t(x-\mu_i) + \ln P(c_i)$$

...ution)

$$= \frac{-1}{2\sigma^2}\left[x^t x - 2\mu_i^t x + \mu_i^t \mu_i\right) + \ln P(c_i)$$

(exp (A) & this eqn are not equal but equivalent when making a decision)

NOTE: our decision favors the prior if x is near the mean

← (equivalent in terms of decision rule)

same for all

$$= \frac{1}{2\sigma^2}\left(2\mu_i^t x - \mu_i^t \mu_i\right) + \ln P(c_i)$$

Since we are looking for discriminative function, which is defining region.

$$g_i(x) = W_i^T x + W_{i0}$$

linear discriminant function
✱ (last page)

Where, $W_i = \frac{1(\mu_i)}{\sigma^2}$, $W_{i0} = ?$ ← can be derived from here

$\to -\frac{\mu_i^T \mu_i}{2\sigma^2} + \ln P(c_i)$

Since $x^t x$ contribute equally/same in all classes hence removed so whatever is depended on (i) is considered

special case
$\Rightarrow W_i = \Sigma_i^{-1}\mu_i$
$W_{i0} = -\frac{1}{2}\mu_i^T\Sigma_i^{-1}\mu_i + \ln P(c_i)$

Page No:
Date: / /

Averaging Strategies for Binary Metrics

NOTE: Micro Precision = Micro Recall = Micro F1 = Accuracy in Multi-class settings

(iii) Weighted Average ← accounts for class imbalance

- Calculate for each class
- take the average weighted by support (no. of true instances) for each class

$$\text{Weighted Avg} = \sum_{i=1}^{C} \left(\frac{S_i}{\sum_{j=1}^{C} S_j} \right) M_i$$

Multi Label Classification Metrics

- Example Based Metrics

(i) Exact Match Ratio (EMR) / Subset Accuracy

- fraction of samples where the predicted label set exactly matches the true label set

$$EMR = \frac{1}{N} \sum_{i=1}^{N} I(y_i = \hat{y}_i)$$

y_i ← set of true labels for instance i; $\hat{y}_i$ ← set of predicted labels for instance i; N ← Total no. of instances

- A prediction is only correct if all labels are perfectly predicted.

$I(\cdot)$: Indicator function (1 if true, 0 if false)

(ii) Hamming Loss

- Lower is better (0 is perfect)
- Similar to no. of labels
- Avg fraction of labels that are incorrectly predicted per instance
- Measure the proportion of the instance-label pairs that are misclassified

$$\text{Hamming Loss} = \frac{1}{N \times K} \sum_{i=1}^{N} |y_i \triangle \hat{y}_i|$$

$\triangle$ ← symmetric difference (XOR operation on the label sets, counting labels in one set but not the other)

$$= \frac{1}{N} \frac{1}{K} \sum_{i=1}^{N} \sum_{j=1}^{K} I(y_i^j \neq \hat{y}_i^j)$$

Page No:
Date: / /

(iii) Jaccard Index / Intersection over Union (I∩U)

- avg jaccard similarity coefficient b/t the true & predicated label sets over all instances

$$\text{Jaccard Index} = \frac{1}{N} \sum_{i=1}^{N} \frac{|y_i \cap \hat{y}_i|}{|y_i \cup \hat{y}_i|}$$

- Range : [0,1]
- penalises both false positives & false negatives in the label sets

Label Based Metrics

- Adapt Binary Metrics by considering each label independently & then averaging
- let :

TP_j, FP_j, FN_j ← for jth label across all instances

(i) Macro Average : Treats all labels equally

$$\text{Macro Precision} = \frac{1}{L} \sum_{j=1}^{L} \frac{TP_j}{TP_j + FP_j}$$

Similarly,

$$\text{Macro Recall} = \frac{1}{L} \sum_{j=1}^{L} \frac{TP_j}{TP_j + FN_j}$$

$$\text{Macro F1} = \frac{1}{L} \sum_{j=1}^{L} \frac{2 \times M.P_j \times M.R_j}{M.P_j + M.R_j}$$

(ii) Micro Average : Weights each instance-label prediction pair equally.

Favors performance on common labels

$$\text{Micro Precision} = \frac{\sum_{j=1}^{L} TP_j}{\sum_{j=1}^{L} (TP_j + FP_j)}$$

$$\text{Micro Recall} = \frac{\sum_{j=1}^{L} TP_j}{\sum_{j=1}^{L} (TP_j + FN_j)}$$

$$\text{Micro F1} = \frac{2 \times \text{MicroPrecision} \times \text{MicroRecall}}{\text{Micro Precision} + \text{Micro Recall}}$$

(iii) Weighted Average

(iv) Sample Average
- calculate metrics for each instance individually, then average

(Considers the performance on each sample equally)

$$\text{Sample Precision} = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i \cap \hat{y}_i|}{|\hat{y}_i|}$$

$$\text{Sample Recall} = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i \cap \hat{y}_i|}{|y_i|}$$

- Handle division by 0 if $|\hat{y}_i| = 0 = |y_i|$
  → Mostly considered 1
- based on F1,

$$\text{Sample F1} = \frac{1}{N}\sum_{i=1}^{N} F1_i$$

Correlation & Aggregation Metrics
- Measure the overall agreement btw predictions & true values.
- Sometimes offering advantages on imbalanced datasets

(i) Pearson Correlation Coefficient (r)
- measures linear correlation between two continuous variables
- less common as a direct classification metric
- Compare predicted probabilities to binary outcomes or feature analysis

for two variables X & Y with N paired samples (x_i, y_i):

$$r = \frac{\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i-\bar{x})^2 \sum_{i=1}^{N}(y_i-\bar{y})^2}} = \frac{\text{Cov}(x,y)}{\sigma_x \sigma_y}$$

↳ $\sigma_x = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})^2}$

where: $\bar{y}, \bar{x}$ → sampled mean

σ_x, σ_y → sampled standard deviations

Range → $[-1, +1]$
- $+1$: perfect +ve linear correlation
- -1: perfect −ve linear
- 0: no linear correlation

Page No:
Date: / /

Aggreemnt : TP×TN
Disaggreemnt : FP×FN

(ii) Matthews Correlation Coefficient (MCC)

- A correlation^bet^ the observed & predicted binary classification
- regarded as a balanced measure which can be used even if the classes are of very different sizes

Ac: Actual
P : Predicted
TP = $\sum A_i P_i$
TN = $\sum (1-A_i)(1-P_i)$
FP = $\sum (1-A_i) P_i$
FN = $\sum A_i (1-P_i)$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Range : [-1, 1]
→ +1 represents a perfect prediction
→ 0 " no better than random prediction
→ -1 " total disagreement bet^ prediction & observation

$A_U = \frac{TP+FN}{N}$

$P_U = \frac{TP+FP}{N}$ · High values of both positive & negative classes to perform good

$\sigma_A = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(A_i - A_U)^2}$ ⇒ $A_U(1-A_U) = \sigma_A^2$

· Advantages :

$\sigma_P = \sqrt{\frac{1}{N}\sum_{i=1}^{n}(P_i - P_U)^2}$ (i) performed over Accuracy & F1 score

(ii) because it requires the model to correctly classify both the majority & minority class to achieve a high score

$\Downarrow$

$\sigma_P^2 = P_U(1-P_U)$ (iii) if model always predict majority class ⇒ MCC → 0 but accuracy is high

(iv) Similarly MCC avoids pitfalls of F1 where high scores can sometimes be achieved by models poor at predicting one class if the other is predicted exceptionally well.

NOTE : Bias Variance Tradeoff says

(i) increasing model complexity generally ↓ bias ↑ variance

(ii) decreasing " " " ↑ bias ↓ variance

$\sum_{i=1}^{N}(x_i - \bar{x})^2$

Page No:
Date: / /

Precision = $\frac{TP}{TP+FP}$

- Use case: when cost of FP is high
  ex: spam detection
  Search engine results (don't want unwanted results at top)
- Limitations: ignore false negatives
  → predict positive only when confident ⇒ missing actual positive instances but high precision
- Exploitation: perfect precision by predicting only 1 instance as positive, provided it's correct

Recall = $\frac{TP}{TP+FN}$

- Use case: when cost of FN is high
  ex: medical diagnosis for serious diseases (don't want to miss person with disease)
  fraud detection (don't want to miss fraud/unauthorized transactions)
- limitations: ignores FP
  → perfect recall by predicting everything pos
  → but this may lead to many FP & low precision
- Exploitation: perfect recall by predicting everything as positive → (1 − FPR)

Specificity (TNR) = $\frac{TN}{TN+FP}$

ability to identify negative class

- Use case: when correctly identifying negatives is crucial
  eg. drug testing (no false accusation of drug use)
- Exploitation: ignores positive class performance (TP, FN).

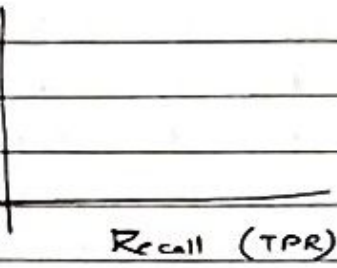
Page No:
Date: / /

Negative Predictive Value (NPV)
(measures accuracy of negative predictions)

$$NPV = \frac{TN}{TN + FN}$$

- use case: When confidence in a negative prediction is needed
  Ex. if a diagnostic test predicts a patient is healthy, NPV tells us the % they are actually healthy
- limitations: can be high even if the model misses many positive cases (high FN), especially if the negative class is large

PR Curve
- As threshold decreases, more instances classified as positive increases Recall but decreases Precision



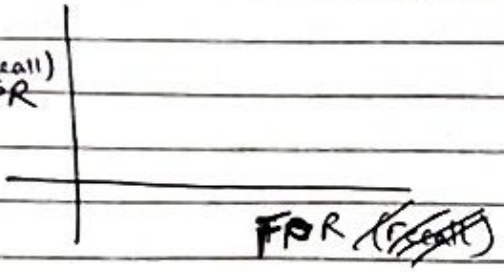
- Ideal Model: curve goes to (0,1) & then straight across to (1,1)
- Random Model: A horizontal line at the level of positive class prevalence (proportion of positive ~~class~~ instances in the dataset)

$$P = \frac{TP}{TP + FP} \approx \frac{Pos}{Pos + neg} = \text{Prevalence}$$

- Good model: Recall shoots the top right corner
- AUC-PR: More informative than AUC-ROC for highly imbalanced datasets where the large no of true negatives can inflate AUC-ROC (area → 1)

(our primary interest is positive class)

ROC curve.
- lowering threshold ⇒ ↑ FPR & ↑ TPR



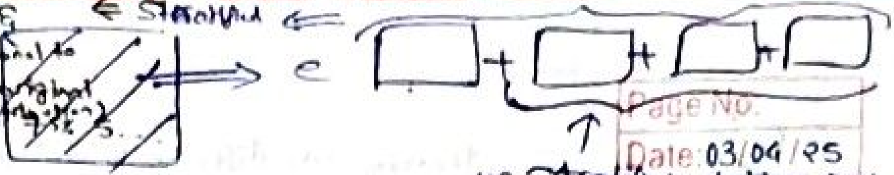
Page No:
Date: / /

- Ideal Model : Curve goes straight upto (0,1) then goes straight across (1,1). Top left corner is best
- Random Model : Diagonal line from (0,0) to (1,1) $\Rightarrow$ TPR = FPR
- Good Model : Bows out towards the top left corner.
- AUCROC : Area Under ROC curve
  - → ranges from 0.5 (random) to 1.0 (perfect)
  - → represents the probability that the model ranks a randomly chosen positive instance higher than a randomly chosen negative instance. Less sensitive to class imbalance than AUC-PR

NOTE !!
(Previously we evaluated the performance of model on training set / how well it is performing on the given set, ⟶
but here we check on unseen data)

(pick from each & keep it proportional to the original distribution) ← Stratified ←

(fig 1)



un Stratified : taking only from 3/4 labels

Date: 03/04/25

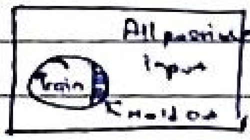
a) How well a classifier do on unseen data?
- model overtrain
- distribution shift : if the test data is very different
- What can we do to measure how the model will perform on an unseen data?

Method 1:

→ Lazy sol^n : (no estimation of performance on unseen data done) ← (CRUX)

this is known as Resubstitution
- only reports on the available data
- full data used
- overestimating performance
- So test is also done on the same data on which it is trained

Method 2: (1 - SHOT)



Keep some part of the data from training

→ Check performance on holdout set

Train 1 model

(A) → if $L_1 \ldots L_m$ found during training, it might give high for some specific labels in case of test } ← unstratified (?) — not good

→ hence go for stratified labels (?) : maintain ratio of labels in training dataset (REFER fig 1)

(# 1-Shot)

giving k models ⇑

(Train same model with different hyperparameters) → (B) NOTE: we can have multiple models on the same training set by changing the hyperparameters etc & check whichever performance is good ⟹ Select & deploy it whichever performs good on testing set ⇑ but this is not good as leakage of data

Why better because when selecting model we didn't consider the holdout set so avoiding leakage of data

(B') {
- Train model ← train set
- Variation of model ← validation set } ← out of M models choose 1
  - ↳ like hyperparameters etc selection
- Test on holdout set ← Test set
  - ↑ gives the estimated performance

Assumption

test set (the holdout) is similar to real world unseen data

K-shot

C

→ training sets different & testing always / different

b) K - Fold Cross Validation

- We create K testing set instead of 1 hold out set to check performance

- K = N : Leave one out method
→ each point used for testing only once

every point is possible to be in test set

(K partition of) Data — Stratified / unstratified

K classifiers — K-1 partition for training / 1 partition for testing

	Test	Train
M_1:	P_1	$P_2 \ldots P_K$
M_2:	P_2	$P_1 P_3 \ldots P_K$

→ best? ⊗ → avg? ⊗ →
→ how to check which best among K
if we want to pick 1/K we compute some metric (precision, accuracy etc)

- K model versions
  → Compute mean
  → risk performance variation

} ← for each model

- helps in determining what performance I can get on unseen data

We compute Confidence Intervals
→ higher confidence interval ⇒ go for that avg mean
What does it tell?
→ This gives sort of some score

- While deploying.
  we train on whole data & not K different version

- Considered as normal distribution :

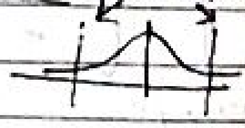
$$\bar{u} \pm Z_{\alpha/2} \frac{S}{\sqrt{K}}$$

mean; normal distribution; 1 - confidence; S ← std deviation; $\sqrt{K}$ ← std error

- This makes us to ask how much area is covered

- NOTE: the reason was I can compute M (precision, accuracy etc) on all K, & take avg but not good

Clip out extreme



- So we go for finding confidence (but on what?)

Rev:

but now → Training size reduced to N but we need test data as well, so we pick the training sample randomly from existing data as (refer fig 2)

(D)

c) Training size in previous one was one less but to maintain training data size ⇒ Bootstrapping

K-shot + Maintain train data size

- Sampling with sampling with replacement done: for creating a dataset of size N
- probability of getting not selected for N rounds of a data point $= \left(1 - \frac{1}{N}\right)^N \approx \frac{1}{e} = 0.36$

→ The unselected points become the test set (i.e. out of bag data)

E_0 : Out of bag data very small ∴ so performance evaluation on it & reporting not efficient

- Points not selected for training data

avg & variance? (*)

$E_{0.632} = 0.36\,(\text{Training performance}) + 0.64\,(E_0)$

↳ final estimated performance

↓ here we go for confidence intervals

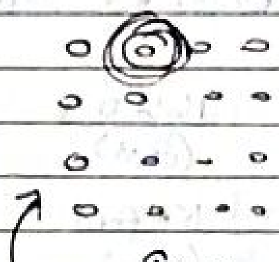


fig 2

Here, one of the points selected by sampling with replacement
⇓
this is bootstrapping

Note:

K-fold cv → no hyperparameter selection (only model performance evaluation)

Note: Bootstrapping is different, we don't consider the testing set over here, the performance is estimated based on above formula*

Page No:
Date: 04/04/25

a) K-NN Classifier

→ data is kept as train hence used → vote on those labels

→ 2 methods: (i) ... (ii) ...

- no training of classifier
- for test, we find top k nearest neighbour & give the label

Adv: no computation cost

Disadv: Compute distance with every point in the data set during testing (Slow prediction) — n distances

for smaller data set (small k): (error = 0) ← k=1 & train = test
⇒ (low bias, high Variance)

- Change of data: no issues
  Change of label: no issues
- Privacy concern: problem — as training data gets exposed
- Optimisation: n distances req (currently)
  we are interested in just k distances
  ex. KD trees (done by)
  ↳ prediction: O(log n) → in normal kd

b) Evaluation Technique (contd)

→ How much the training data to is deprived of?

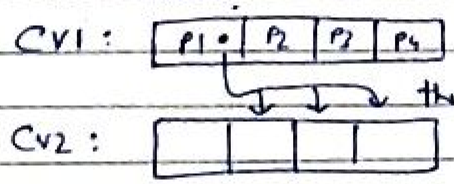
	Resampling	Bias	Var
Hold Out (fast)	X	High	high
K-fold (slow)	Not repeated	Moderate	mod
Bootstrap (Slowest)	Repetition	Higher (E_0) ↓ Moderate ($E_{0.632}$)	low

→ Best of CV & Bootstraping

c) Repeated CV: previously CV was done only 1 time

→ start with fresh set of data ∴ hence no repetition
→ large no. of tries
→ How? ← 4 instances of model

give performance estimation



the previous point which belonged to P1 now can go to either of P_2, P_3, P_4 for CV2

⇒ for each CV, we get a performance score
⇒ compute average → we get a stable performance estimate

Page No:
Date: / /

: (1) treat them as 1 set & get the maj (smly)
(2) Weighting: first K-NN
gets more weight

→ Best hyperparameter

gives ~~both~~ the hyperparameter
~~as well as performance~~ (by doing something)

d) Hyperparameter Tuning

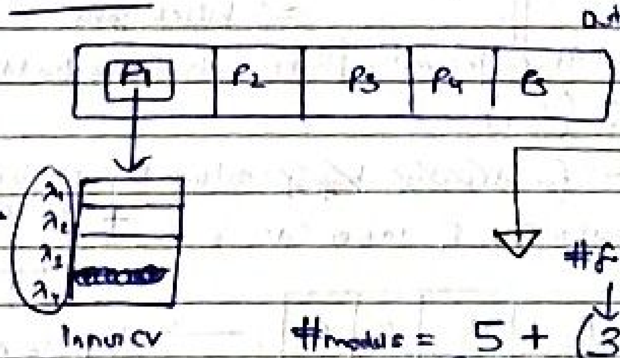
- let's do 3 fold CV → folds should be constant here
  λ : 0.2   0.7   (6 models, 6 performance for hyperparameter estimate get final estimate)

CV1 : $\lambda = 0.2$ → m_1 m_2 m_3 → Average
CV2 : $\lambda = 0.7$ → m_1' m_2' m_3' → Average
} whichever is best go with it
⇓ but not good (as overestimation happens)

- Model with 4 hyperparameters configs
  ⇓
  5 fold CV
  # models = 5×4 + 1
  5×4 → for deciding the hyperparameter
  1 → for training the final model on whole data set
  → how many times do we run the model here? (~~5~~ times?)

Variance of Nested CV : Done for hyperparameter tuning + performance estimation

- Outer CV → Inner CV {basic structure}



← ex. 5 fold outer
{ 3 fold inner
{ 4 hyper-parameter

(for each λ we run the Inner CV)

for each outer

#models = 5 + (3 × 4) × 5 + (12) + 1
(5 → for outer; 3 → #folds; 4 → # hyperparameters)

M_1 : Test : P_1 , Train : P_2 P_3 P_4 P_5 → for one particular setting of λ (the λ_1)
↓
Make 3 partitions of these

Create M_1' : Test : D_1 Train : D_2 D_3
M_1'' : Test : D_2 Train : D_1 D_3
M_1''' : Test : D_3 " : D_1 D_2.

repeat for λ_i $(i \neq j)$
↓
get the avg value ε_i

then go for next fold i.e. P_i $(i \neq j)$ ← repeat λ for P_i

Page No:
Date: / /

- estimation of performance done on different data set than the training set ⇑ here it's an advantage

previously 1 fold: it was overlapping

- nested CV → more reliable estimation of model performance

models: 5 + 60 + (12) + 1

↑ hyperparameter setting

---

c) Which model to select?

Model A: 2 hyperparameter configuration

" B: 3 " "

Simple CV with 5 fold: A : 5×2+1 = 11 (used for deciding performance)

B : 5×3+1 = 16

⇒ Which gives an idea on which is better model

(We need to select one among them)

Do we have all 4 hyperparameters {setting} before only? (Yes)

Nested CV — for estimating performance + to find out which hyperparameter

→ consists of outer CV & inner CV

Class example:

P_1	P_2	P_3	P_4	P_5

→ (i) Data partitioned into 5 sets

(ii) because we go for 5 fold outer CV

⇓ to determine out of 4 hyperparameters:

~~the CV~~
~~CV~~
~~CV~~
~~CV~~

now since each CV_i will run an inner CV of 3-fold with 4 different hyperparameter settings

let's say: We are doing CV_1

Test set : P_1

Train set : $P_2 + P_3 + P_4 + P_5$

\# for CV1

What we do is?

⇒ We divide $P_2+P_3+P_4+P_5$ into 3 partitions as 3 fold inner CV

D_1	D_2	D_3

→ run this using $\lambda_1, \lambda_2, \lambda_3, \lambda_4$

This gives us 3×4 models ← → for each λ, it will be repeated 3 times

Test	Train
D_1	D_2 D_3
D_2	D_1 D_3
D_3	D_1 D_2

→ plus a good hyperparameter setting

next with that hyperparameter ← setting, we train a new model on $P_2+P_3+P_4+P_5$

this new trained model ← is tested on P_1

Which gives us the performance

→ now this is repeated for the remaining possible partitions out of the 5 → & then take average

Hence we got: $(3 \times 4 + 1) \times 5$

↑ for hyper parameters

↑ for final model on the best hyperparameter

$(2 \times 7 + 1)^4$

= 60

now, $(3 \times 4 + 1)$ was extra:

→ so prev one gave us performance estimation

→ how to select final hyperparameter & get the model

→ we do 3 fold final hyperparameter selection (given in Q)

Which is best?

Initial partition: | P_1 | P_2 | P_3 | P_4 | P_5 | ⇒ | D_1 | D_2 | D_3 | ⇒ 3 fold CV on 4 different λ setting

train on full data set ← best λ ← λ setting

⟹ gives the final trained model

Page No:
Date: / /

→ Hitesh Arora – Amazon
  → clean data most of the time
  → data labelling etc (Imp)
  → different stakeholders
- (i) Model – in research & production (diff)
(ii) Amazon Scout – requirements (LIDAR, camera)
(iii) LLM Model – Agentic workflows
  • Agents existed from beginning
  • Use of Agents to LLM

→ Not perfect model, it's reliability
  • like e we don't focus on accuracy much (but in research & college, yes)
→ Amazon Scout (used transformers (check))
→ Development with LLM model (Try)
→ Deep learning, V/s RL V/s LLM V/s Where does each hold?
  ↳ can do classification as well?
→ ~~KSG~~ ? (or AGI is) → Artificial General Intelligence
→ Mental model on code base – no AI able to give everything: (Code Review still done)
  → So reading code is still required for development
  → Hence having the skill is (must)
→ Deep learning resources? → (i) Fundamental of models
  ← Prof, Courses online
(ii) For new models to understand → Papers, AI tools (!!), Langchain Langgraph (tools)
(iii) Projects (Small ones)

Page No:
Date: 07/05/25

→ The crux of it is using probability to make classification decisions ↓ Decision Rule

Bayesian Classifier
Min Error Classification — arg max $P(C_i|x)$
min Risk " — arg min $R(\alpha_i|x)$

$$P(C_i|x) \propto P(x|C_i)\,P(C_i)$$

This sort of classifier is called generative Classifier

- Class conditional density either given or can be estimated ⇒ Since data points can also be generated ← Why is it called that?
- not just classifier sample
- but data point/density can be computed

{Generative, Discriminative classifier / Discriminitative}

⇒ We can get another classifier based on discriminant function : let's say $g_i(x)$: D.F

Decision rule : arg max $g_i(x)$

⇒ Binary classification : $g(x) = g_1(x) - g_2(x)$

⇒ We can get discriminant function in various ways, one of the ways is ↑

if → $g_i(x) = \ln p(x|C_i) + \ln p(C_i)$

Derive it if → $p(x|C_i) \sim N(\mu_i, \Sigma_i)$ like for each cases

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t \Sigma_i^{-1} (x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i|$$

(features have same variance & linearly independent)

Case I: $\Sigma_i = \sigma I \quad \forall i$

Each of these belong to a region/class

This is derived from the Probabilistic classifier but simpler

$$g_i(x) = w_i^t x + w_{i0}$$

— linear classifier (or discriminant function)

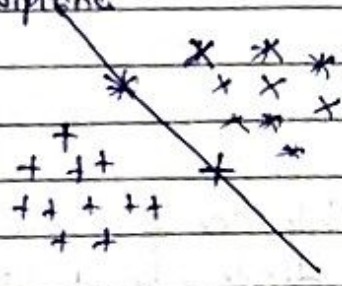
$$\rightarrow w_i = \frac{\mu_i}{\sigma^2}$$

$$\rightarrow w_{i0} = \frac{-1}{2\sigma^2} \mu_i^t \mu_i + \ln P(C_i)$$

Also known as discriminant Classifier (why? .....)

Decision Boundary → Same as separating hyperplane

$$\hookrightarrow g_i(x) = g_j(x)$$



$$w_i^t x + w_{i0} = w_j^t x + w_{j0}$$

Algebraic manipulation

$$\Leftarrow\!\Rightarrow \boxed{w^t (x - x_0) = 0}$$

it's a separating hyperplane passing through x_0 & normal to w

NOTE: vectors are considered as, $x \in \mathbb{R}^{d \times 1}$

Where $w = \mu_i - \mu_j$: hyperplane separating the two regions R_i & R_j

$$x_0 = \frac{1}{2}(\mu_i - \mu_j) - \frac{\sigma^2}{\|\mu_i - \mu_j\|^2} \ln \frac{P(C_i)}{P(C_j)} (\mu_i - \mu_j)$$

Special case

Case 1a: if $P(C_i) = P(C_j)$, the separating hyperplane would be the bisector of means (??)

→ x_0 : midpt of line joining μ_i & μ_j

→ Separating hyperplane ~~would~~ is $\perp^r$ bisector of the line joining means

Case 1b: $P(c_i) \neq P(c_j)$ (far from mean common class mean)

⇒ The separating hyperplane is shifting away ^ (#Interpret separating hyperplane)

#Assumption) ⇒ To either of the class

↳ Which class? like if $P(c_i) < P(c_j)$ or $P(c_i) > P(c_j)$

Class conditional density: $N(\_,\_)$, $\Sigma_i = \sigma I$

Case 1c: $P(c_i) = P(c_j) \quad \forall i,j)$

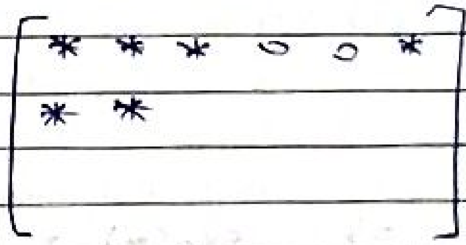
like if more than 2 classes ⇒ they are equiprobable

Verify : $g_i(x) \propto -\|x - \mu_i\|^2$

⇒ ToDos:

Same for all classes

Case II : $\Sigma_i = \Sigma$



(i) Write down $g_i(x)$ — ?

(ii) What happens $P(c_i) = P(c_j) \quad \forall i,j$

(iii) Do we get linear discriminant function?

(iv) What is the form of decision boundary / separating hyperplane?

Case III : Σ_i = arbitrary ← different for each class

↑ This is of the quadratic discriminant form — (Verify it)

Previous cases were continuous, but now, what if features are discrete?

x_i : taking one of 1 to m_i discrete values

Spl Case : Assumptions:
- Binary features — x_i < 0, 1
- Features are conditionally independent

⇒ It is a binary classifier

$p_i = P(x_i = 1 | c_1)$

$q_i = P(x_i = 1 | c_2)$

$$P(C_1|x) \propto p(x|C_1)\,P(C_1)$$

Page No:
Date: / /

→ features are distinctly independent given the class (assumption)

$$p(x_1, \ldots, x_d|C_1) = \prod_{i=1}^{d} p(x_i|C_1)$$

$$= \prod_{i=1}^{d} p_i^{x_i}(1-p_i)^{1-x_i}$$

↑ if $x_i = 1$, $(1-p_i)$ goes away

(Bernoulli)

$$p(x_1 \ldots x_d|C_2) = \prod_{i=1}^{d} q(x_i|C_2)$$

$$= \prod_{i=1}^{d} q_i^{x_i}(1-q_i)^{1-x_i}$$

$$\Rightarrow g_0(x) = \ln \frac{P(x|C_1)}{P(x|C_2)} + \ln \frac{P(C_1)}{P(C_2)}$$

↑ in this particular case,

Verify: $g(x) = \sum_{i=1}^{d} w_i x_i + w_0$

→ $g(x) > 0 \rightarrow$ class 1

→ $g(x) < 0 \rightarrow$ class 2

where,

Verify ←

$$w_i = \ln\left(\frac{p_i(1-q_i)}{q_i(1-p_i)}\right)$$

$$w_0 = \sum \ln\left(\frac{1-p_i}{1-q_i}\right) + \ln\left(\frac{P(C_1)}{P(C_2)}\right)$$

Page No:
Date 05/05/25

How do we objectively compare different regions?
like Bangalore & San Francisco
↓ capture semantic / structural similarity

Step1 ## Embedding : creation of numerical vectors using objects / words
↑ ↓ words are related with close ones (semantic)
Created by word2vec
→ Why? — because can't be represented in form of numerical or categorical data
a technique in NLP for obtaining vector representations of words

Step2 How do we represent a region?
Why do we want to do it? — like finding a region where cost is high/low, where to install next McDonalds store?
→ What sorts of feature will you take?
ex. proximity to the places, demographics, nice water body or park
Dataset
⇓
how to convert these so ML models can understand? → embeddings

→ for discussion: 2 datasets considered
since dynamic ← (i) Mobility Dataset
(ii) Satellite Images — Similar to facial embeddings
data, how is it handled?

Page No:
Date: 11 /04/25

Causal ML

Correlation, Causation, Fallacy
↓
doesn't tell everything
↓
is all about directionality → If A & B are correlated, Which cause which?
a) Could be: A → B
b) or: B → A
c) or: C causes both A & B

Simpson's Paradox

How do we know when to stop or these are enough features for our model?
- Random(ised) control test → something like we test one group with something & others with another & check the output

not good for cases like to detect if smoking causes cancer

Causal Model - Potential Model → every person has 2 potential outcomes
y_1 : if taken (.)
y_0 : if not taken

Potential outcome independent of the assignment (smtg)

Frisch Waugh Lovell Theorem - a way to isolate the effect of one variable in a multiple

Summary

1) Correlation ≠ Causation → one variable directly affects the other
↓
- 2 var are correlated if they move together.
- it shows association but not cause

2) ML models often learn patterns (correlations) but that's not enough when we want to intervene or understand why something happens.

Page No:
Date: / /

3) ~~If~~ A Simpson's Paradox

- A trend appears in several subgroups of data but reverses when the groups are combined.

Ex.	Group	Treatment Success	Control Success
	Men	90%	80%
	Women	70%	60%
	Combined	75%	85%

- Why? → Maybe more Women (with lower success) are in treatment group overall. (confounder)
- Always consider hidden variable when analysing data.

4) Randomised Controlled Trials

- A way to test causality by randomly assigning subjects to: (i) Treatment group
  (ii) Control group

Why randomisation? - remove bias & confounding

We can only observe one of them at a time

Linear regression

Page No:
Date: 17/04/25

$$\arg\max \underbrace{p(c_j|x)}_{\uparrow} \propto p(x|c_j)\,p(c_j)$$

To find this: $p(x|c_j) \sim f(x, \theta)$

$N(\underbrace{\mu_j, \Sigma_j}_{\text{known}})$

Now,

let's consider: $p(x|c_j) \sim N(\mu_j, \Sigma_j)$ ↙

What if μ_j unknown & Σ_j known
if μ & Σ_j unknown
$p(c_j) \sim$ unknown

Q How do you estimate the parameters?

Parameter Estimation

- (a) Maximum Likelihood estimation (MLE) ↑ (our course is focused on this)
- (b) Bayesian Estimation

how does both differ?

(i) MLE assumes parameter is fixed

(ii) Bayesian estimator considers it as random variable with known prior

→ Try to find a parameter value which gives maximum probability for the observed/given data

Observation/data leads to posterior density/probability

Maximum Likelihood Estimator

- General setting

$$D = \left\{ (x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)}) \right\}$$ ↖ indices of samples

$$y^i \in \{c_1, c_2 \ldots c_K\}$$

• How do we use MLE ?

→ data divided into K subsets : $D_1, D_2 \dots D_K$ ← Samples belonging to class 1

→ samples in D_j are i.i.d (independent & identically distributed) — (P?)

→ $P(x|C_j)$ is assumed to be coming from parametric form ($\sim F(x, \Theta_j)$) ← random variable

(particular sample belonging to a class)

→ Θ_i, Θ_j : functionally independent $i \neq j$

↑ parameters ← if this is known we can have our classifier computed by this way

• What is the likelihood function ?

$L(\Theta_j | D_j)$ ← in general (Θ), not considered index j

parameter Θ_j (for) D_j subset (doubtful)

$L(\Theta_j, D_j) := P(D_j | \Theta_j)$

joint probability distribution $= P(x_{D_j}^{(1)}, x_{D_j}^{(2)}, \dots x_{D_j}^{(n_j)} | \Theta_j)$

→ since it is i.i.d, therefore

$= \prod_{i=1}^{n_j} P(x_{D_j}^{(i)} | \Theta_j)$

AIM: estimate the parameters Θ_j of each class conditional densities independently

$L(\Theta, D) = \prod_{i=1}^{n} P(x^{(i)} | \Theta)$ ← In general we say this but for each subset we define

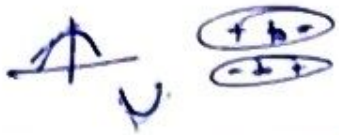
$$\hat{\Theta}_j = \underset{\Theta_j}{\operatorname{argmax}} \prod_{i=1}^{n_j} P(x_{D_j}^{(i)} | \Theta_j)$$

• Generally we consider log likelihood

$$l(\ ) = \sum_{i=1}^{n_j} \ln P(x_{D_j}^{i} | \Theta_j)$$

$$\Rightarrow \hat{\Theta}_j = \underset{}{\operatorname{argmax}} \prod_{i=1}^{n_j} P(x_{D_j}^{i} | \Theta_j) = \underset{}{\operatorname{argmax}} \sum_{i=1}^{n_j} \ln P(x_{D_j}^{i} | \Theta_j)$$

ToDo : Just check how it came



Page No:
Date: / /

• Univariate Normal Distribution

$$P(x_k \mid \underbrace{\mu, \sigma^2}_{\text{(unknown)}}) = \frac{1}{(\sqrt{2\pi})\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$

$$\ln\left(P(x_k \mid \mu, \sigma^2)\right) = -\ln \underbrace{(2\pi)^{1/2}\sigma}_{(2\pi\sigma^2)^{1/2}} - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$$

That's why

$$\ell(\mu, \sigma^2) = \sum_k \left(-\frac{1}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(x_k-\mu)^2\right)$$

$$\frac{\partial \ell}{\partial \mu} = 0 \quad ; \quad \frac{\partial \ell}{\partial \sigma^2} = 0$$

verify: $\boxed{\hat{\mu} = \frac{1}{n}\sum_k x_k}$ ← Sampled mean

$\boxed{\hat{\sigma}^2 = \frac{1}{n}\sum_k (x_k - \hat{\mu})^2}$ ← Sampled variance

Exercise (for next class)

for posterior pb → $\boxed{P(c_j \mid x) = P(x \mid c_j)\, P(c_j)}$ — (A)

$P(x \mid c_j) \sim \mathcal{N}(\mu_j, \Sigma_j)$

multivariate normal dist^n

Step 1:

req μ_j Calculation { Case 1: μ_j — unknown, $\Sigma_j = \Sigma$ $\forall j$ known

both μ_j & Σ_j { Case 2: μ_j, Σ_j — unknown

$P(y=1) = q$, $P(y=0) = 1-q$ $\boxed{q\,?}$

⟹ Then we can find (A)

→ As in to find the # of the parameters

→ Parameter Estimation
(10 marks Q)

Page No:
Date: 21/04/25

$$D = \{(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)})\}$$

- $x^{(i)} \in \mathbb{R}^d$, $y^{(i)} \in \{0, 1\}$

# of parameters
$(d^2 + d + d + 1)_{+q}$
↑ from Σ ↑ from μ_1 ↑ from μ_0

Assumptions:

(i) $p(x|y=0) \sim N(\mu_0, \Sigma)$ ← Unknown

(ii) $p(x|y=1) \sim N(\mu_1, \Sigma)$ ← unknown

(iii) $p(y=1) = q$ ← unknown $q \in \mathbb{R}$

$p(y=0) = 1 - q$

$\mu_0 \in \mathbb{R}^d$	$\Sigma \in \mathbb{R}^{d \times d}$
$\mu_1 \in \mathbb{R}^d$	q

How to use Maximum Likelihood concept to find the parameters

$$\Theta = \{\mu_0, \mu_1, \Sigma, q\}$$

Likelihood

$$L(\Theta|D) = p(D, \Theta)$$

$$= p((x^{(1)}, y^{(1)}) \ldots (x^n, y^n), \Theta)$$

→ because of independence

$$= \prod_{i=1}^{n} p((x^{(i)}, y^{(i)}), \Theta)$$

$$= \prod_{i=1}^{n} p(x^{(i)} | y^{(i)}, \Theta)\, p(y^{(i)}, \Theta)$$

← To make it

Simpler, take log-likelihood ∵ As it is non decreasing function
⇒ maximizing $\ell(\Theta, D) = L(\Theta, D)$

$$\ell(\Theta|D) = \sum_{i=1}^{n} \ln p(x^{(i)} | y^{(i)}, \mu_0, \mu_1, \Sigma, q)$$

$$+ \sum_{i=1}^{n} \ln p(y^{(i)}, \mu_0, \mu_1, \Sigma, q)$$

Page No:
Date: / /

$$\theta^* = \arg\max_{\theta} L(\theta|D) = \arg\max_{\theta} \ell(\theta|D)$$

NOTE: q doesn't make any impact for

first term $\xi(\mu_0, \mu_1, \Sigma)$ doesn't effect second term

$\Rightarrow$ hence, it is removed in $\ell(\theta|D)$

$$\ell(\theta|D) = \sum_{i=1}^{n} \ln\left(p(x^{(i)}|y^{(i)}, \mu_0, \mu_1, \Sigma)\right) + \sum_{i=1}^{n} \ln\left(p(y^{(i)}, q)\right)$$

$$p(x|y=0) = \frac{1}{(2\pi)^{d/2}(\Sigma)^{1/2}} \underbrace{\exp(\dots)}_{f_0(\mu_0, \Sigma)}$$

$$p(x|y=1) = \frac{1}{(2\pi)^{d/2}(\Sigma)^{1/2}} f_1(\mu_1, \Sigma)$$

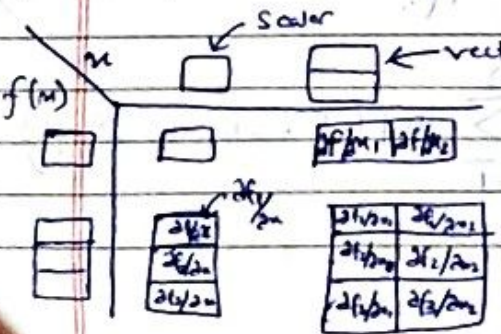
$$p(y) = q^y (1-q)^{1-y}$$

- Now, take gradient

$$\nabla\ell(\theta|D) = 0 \quad \text{\# (Calculate this)}$$

$$\Rightarrow \frac{\partial\ell}{\partial\mu_0} = 0, \quad \frac{\partial\ell}{\partial\mu_1} = 0, \quad \frac{\partial\ell}{\partial\Sigma} = 0, \quad \frac{\partial\ell}{\partial q} = 0$$

- How to calculate when vectors involving $\leftarrow$ like matrices
  - $\rightarrow$ Note: log-likelihood is scalar



Book
Maths for ML
→ Foundational Maths for revision

Page No.
Date / /

M: matrix $d_1 \times d_2$ (# expressions will be given in exam if at all asked for derivatives)

i) $\frac{\partial M}{\partial x} = \begin{pmatrix} \frac{\partial M_{11}}{\partial x} & \cdots & \frac{\partial M_{1d_2}}{\partial x} \\ \vdots & & \vdots \\ \frac{\partial M_{d_1 1}}{\partial x} & \cdots & \frac{\partial M_{d_1 d_2}}{\partial x} \end{pmatrix}$

($\frac{\partial M}{\partial x}$: Scalar)

ii) $\frac{\partial M^{-1}}{\partial x} = M^{-1} \frac{\partial M}{\partial x} M^{-1}$

iii) $\frac{\partial [Mx]}{\partial x} = M$

iv) $\frac{\partial}{\partial x}\left[y^t x\right] = \frac{\partial}{\partial x}\left(x^t y\right)$

$= y$ (here, y is independent vector of x)

v) $\frac{\partial}{\partial x}\left(x^T M x\right) = \left(M + M^T\right) x$

vi) $\frac{\partial \log |\Sigma|}{\partial \Sigma} = \Sigma^{-1}$

vii) $\frac{\partial}{\partial \Sigma}\left(x^T \Sigma x\right) = -\Sigma^{-1} x x^T \Sigma^{-1}$

Verify the below:

i) $\phi = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\left(y^{(i)} = 1\right)$ ← indicator fn (it tells sum up all samples which belong to class 1)

ii) $\mu_0 = \frac{\sum_{i=1}^{n} x^{(i)} \mathbb{1}\left(y^{(i)} = 0\right)}{\sum_{i=1}^{n} \mathbb{1}\left(y^{(i)} = 0\right)}$ ← This is sample mean

here features are comes from continous distribution & have gaussian distribution

A_0 : i st $y_i = 0$

A_1 : i s t $y_i = 1$

Page No:
Date: / /

(iii) $\Sigma = \frac{1}{n}\left[\sum_{i \in A_0} (x^{(i)} - \mu_0)(x^{(i)} - \mu_0)^T + \sum_{i \in A_1} (x^{(i)} - \mu_1)(x^{(i)} - \mu_1)^T\right]$

Case 1 { # it is also possible that (n) features are coming from discrete random variables & then estimate parameters → (of ?)
- here distribution normal or not — (check)
hence find parameters

Probabilistic interpretation of Linear Regression (the note contains this)

Case 1 → (Binomial distribution)
(If nothing is given, take derivative of $P(H=61|p)$, & then test on all possibilities of p) *** * if p-value < 0.05 $\Rightarrow$ H_0 (null hypothesis) is rejected & the model is statistically significant

example. An unfair coin is flipped 100 times, and 61 heads are observed. The coin either has probability $\frac{1}{3}$, $\frac{1}{2}$, or $\frac{2}{3}$ of flipping a head each time it is flipped. Which of the three is the MLE ?

→ pmf : $P(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $0 \le p \le 1$

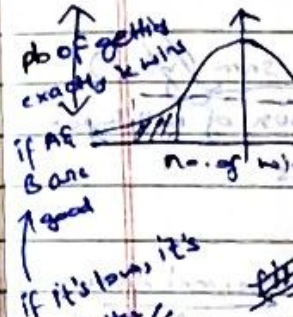
$x = 0, 1, 2, \ldots, n$

given, $n = 100$, we need to find which (p)

$P(H=61 \mid p=\frac{1}{3}) = \binom{100}{61}\left(\frac{1}{3}\right)^{61}\left(\frac{2}{3}\right)^{39} = 9.6 \times 10^{-9}$

$P(H=61 \mid p(\frac{1}{2}) = 0.007$

$\boxed{P(H=61 \mid p=\frac{2}{3}) = 0.040}$ ✓ ← maximum hence MLE is $p = \frac{2}{3}$ //



→ What is normal & binomial distribution?

Page No.
Date 23/04/25

* it starts with some assumption & computes p-value (that's how likely is this assumption)

$(\mu^{(1)} - \mu_1)^T$

Statistical Significance Testing

(i) ~~many~~ Null hypothesis : default assumption (that there is no effect or no difference)

(ii) different methods exist for that

Comparing 2 models

(a) Naive Solution

Win Ratio: K_A → (Probability of A winning)

Compare A & B → K

- How significant is the test?

Classifies K no. of times ($K_A \geq K_B$)

K_A ↑ win times of A

K_B ↑ win times of B

⇒ Why this method is wrong? / (bad)

- magnitude of beat
- no. of test performed : same ratio for diff no. of test can be achieved

ex. 0.8 ← for n_1 small or n_2 large

Hence we go for: Win Ratio + No. of trials

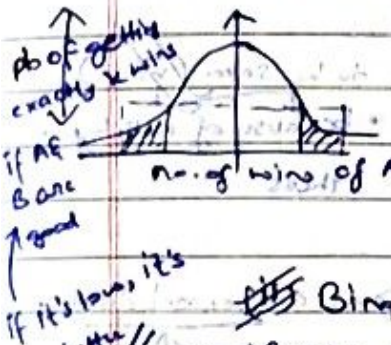
(1) Sign Test → (No. of wins)

→ how many times comparisons done } Assumptions

→ how many time A has beated B

⇓

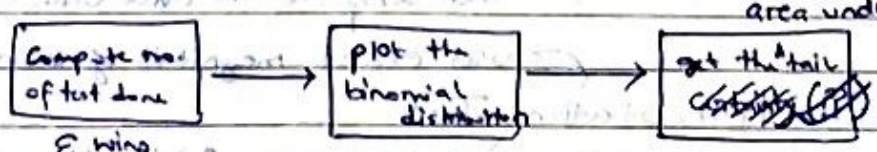
larger the no. of wins
Smaller area under the curve
⇒ more certain that A beat B



~~(1)~~ Binomial Test

→ p value tells what is the chance of observing current data when Null hypothesis is true

→ lower pvalue is good



→ Final conclusion : pvalue (<0.05) : there is a meaningful difference b/w the performance of A & B

for sign test

gives area under the tail

(more than 530 wins good)

(each win is not the same) → Fixed reward based on rank

(ii) Wilcoxon Signed Rank
→ here we consider the magnitude of beaten as well

Performances	A	B	Diff
T1	0.7	0.73	-0.03
T2	0.8	0.75	0.05
T3	0.88	0.81	0.07
T4	0.35	0.39	-0.04

Step 1: Arrange differences in the ascending order (ignore sign)
Step 2: Put back the sign
Step 3: Replace each entry by rank along with sign
Step 4: Sum all positive values (W^+) & negative values (W^-)
Step 5: Choose the smaller value as W

Step 1: Arrange the diff in ascending order (ignoring signs)

~~T4 → T1 →~~
$T_1 \rightarrow T_4 \rightarrow T_2 \rightarrow T_3$
-1 -2 +3 +4

Step 2: → rank them

Step 3: Club together -ve & +ve values
W^- W^+

Why? —— Step 4: Choose the smaller value as W

• Smaller the value more extreme observation
• farther away from null hypothesis

⇒ Both W^- & W^+ should come out to be same (why?)
• because of null hypothesis

Then plot it ⇒ gives a normal distribution curve (diff than one was binomial)
(i.e frequency of that score)
⇓
then compute the area under curve

↗ uses exact magnitude of wins

(iii) Permutation Test
(a) Compute diff of performance : then take sum & avg of differences
(b) Do 2^m (at max) change (i.e A, B value swapped)
↗ for m comparisons
(how?)

Page No:
Date: / /

greater or less than for 1 tailed test

(c) After each permutation, we get diff avg

(d) Next find out, which avgs are equally or more extreme than original observation (ignore the sign)

This gives, p value

greater or less than both considered for two tailed tests

(e) No distribution assumption (like prev case)

(Uncertainty == area under the tail)

(N) Paired T-Test : Assumes normal distribution → follows t-distribution ≈ similar to with heavier tails

→ $d_1, d_2, d_3 \dots d_k$ } ← diffrns observed in performances

→ Find mean for (dmean)

→ find std deviation

Compare the mean difference between 2 related groups (tests) if the mean diff B statistically significantly different from zero

$$t = \frac{d_{mean}}{S_d/\sqrt{k}}$$

How many std errors is my observation away from 0

where d_{mean} = Sampled mean

S_d = Sample variances

k = Sample sizes

Use the normal distribution to find p-value

- If noisy ⇒ use Sign or Wilcoxon's
  If clean data ⇒ use paired t-test

- If p value very low → it tells A will definitely ^ B beat but not by how much?

⇓

told by Cohen's d = $\frac{d_{mean}}{S_d}$

(it tells practical importance)

Statistical significance ≠ Practical Importance

X X

(End of syllabus)

t-distribution: low DF k ⇒ heavier tail

higher k ⇒ lighter tail

∞ k ⇒ normal distribution

Page No:
Date: 01/04/25

- Bias Variance Trade Off
- Performance Measures
- Performance Estimation — on unseen data
- Statistical Significance Testing

} Post-Mid

- Clustering
  - K means variants : Quality of seeds imp.
  - Klemsung's Theorem
  - Anticlustering
  - Evaluation Metric
    - Intrinsic
    - Extrinsic

trick: don't look : KNN
the initial K - nearest neighbors

PreMid Sem

- Regression

ANN - Data reg to be in numerical format

DBSCan

Hierarchical Clustering

---

* (last page contd) (DIY)

→ Decision surface for any linear machine classes such that, the hyperplane is defined by,

$g_i(x) = g_j(x)$ for bi categorical with the highest posterior probabilities

where $g_i(x) = w_i^T(x) + w_{i0}$

$\Rightarrow w^t(x - x_0) = 0$

$w = \mu_i - \mu_j$

$w_i^T x + w_{i0}$

$= w_j^T x + w_{j0}$

$(w_i^T - w_j^T)x = w_{i0} - w_{j0}$

$(w_i - w_j)^T x = w_{i0} - w_{j0}$

$(w_i - w_j)^T x = (w_i - w_j)^T x_0$

$w^T(x - x_0) = 0$

$w = \mu_i - \mu_j$

(**)

$x_0 = \frac{1}{2}(\mu_i + \mu_j) - \frac{\sigma^2}{\|\mu_i - \mu_j\|^2} \ln \frac{P(w_i)}{P(w_j)} (\mu_i - \mu_j)$

(found using $P(w_i|x) \propto P(w_j|x)$ * & then taking log & removing terms)

Page No:
Date: / /

Case 2: $\Sigma_i = \Sigma$

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t \Sigma^{-1}(x-\mu_i) + \ln P(\omega_i) \quad \leftarrow c_i$$

If $P(\omega_i) \rightarrow$ same for all classes, decision rule remains that choose the feature vector x to the class with the nearest mean vector of class c

if $P(\omega_i) \rightarrow$ biased : the decision is in favor of the prior more likely class

expand : $(x-\mu_i)^t \Sigma^{-1}(x-\mu_i)$

$\Rightarrow x^t \Sigma^{-1} x$ ignore as it is independent of i

$$g_i(x) = w_i^t x + w_{i0}$$
$$w_i = \Sigma^{-1}\mu_i$$
$$w_{i0} = -\frac{1}{2}\mu_i^t \Sigma^{-1}\mu_i + \ln P(\omega_i)$$

Since the discriminant function is linear, the outcome of the decision boundaries are again, the hyperplanes with the equation:

$$w^t(x-x_0) = 0$$
$$w = \Sigma^{-1}(\mu_i - \mu_j)$$

$$x_0 = \frac{1}{2}(\mu_i + \mu_j) - \left\{ \ln\left[\frac{P(\omega_i)}{P(\omega_j)}\right] \Big/ (\mu_i - \mu_j)^T \Sigma^{-1}(\mu_i - \mu_j) \right\} \times (\mu_i - \mu_j)$$

$\Rightarrow$ The difference this hyperplane has as compared from the one in case 1, is that it is not orthogonal to the line between the means & also it doesn't intersect halfway the point between the means if the priors are equal.

WHAT ARE WE DOING? → P( ) → Gaussian Models (estimate)

↓

Decision rule & Compute discriminant fn

↓

then find the decision boundary

↓

classify New point

Case 3: Σ_i = arbitrary

⇒ only term that is dropped is $\frac{d}{2}\ln 2\pi$

⇒ Hence the discriminant function is no more linear, it inherently stays quadratic

$$g_i(x) = x^t W_i x + w_i^t x + w_{i0}$$

Where, $W_i = -\frac{1}{2}\Sigma^{-1}$

$$w_i = \Sigma_i^{-1}\mu_i$$

$$w_{i0} = -\frac{1}{2}\mu_i^t \Sigma_i^{-1}\mu_i - \frac{1}{2}\ln|\Sigma_i| + \ln P(w_i)$$

⇒ decision surfaces are hyperquadrics

i.e general form of hyperplanes, pair of hyperplanes, hyperspheres, hyperparaboloids, and more freaky shapes

$\Sigma^{-1}\mu_i - \mu_i^t = 0$

NOTE: Discriminant function derived only by ** substituting P( ) term & then removed to linear or quadratic form

Distributions

a) Binomial : $\binom{n}{x} p^x (1-p)^{n-x}$

b) Bernoulli : $p^{x_i}(1-p)^{1-x_i}$ where $x_i = 1$ or 0

↓ take logarithm

depending upon what's selected

i.e $L = \prod p^{x_i}(1-p)^{1-x_i}$

$$\log L = \sum x_i \log p + \sum (1-x_i)\log(1-p)$$

We are finding the mathematical condition under which a new input should be classified into a particular group, based on probability models for each group.

Page No:
Date: / /

NOTE: # Statistial Significance Test

a) $p(value) = P(data \mid H_0)$

→ pvalue $\neq P(H_1)$

pvalue $\neq P(H_0 \mid data)$

→ pvalue $\neq P(H_0 = false)$

$P(H_0 \mid data) \cdot P(data) = P(data \mid H_0) P(H_0)$

b) pvalue is area under the tails

c) Wilcoxon Signed Rank

Building Wilcoxon Test for small n

- N comparisons → 2^N possible sign combination of ranks
- W value Close to 0 : Reject H_0
- H_0 : if A & B are similar in performance then positive & negative ranks will be alternative

- for high value of N : Approximation with normal distribution
→ Consider only W^+ : Range : 0 to $\frac{N(N+1)}{2}$

$\mu = \frac{N(N+1)}{4}$: Half the ranks are positive

$\sigma = \sqrt{\frac{n(n+1)(2n+1)}{24}}$

$Z = \frac{W^+ - \mu}{\sigma}$: How many standard deviation away from mean

$|Z| \geq 1.96 \rightarrow$ pvalue ≤ 0.05

Page No:
Date: / /

Sign →	Wilcoxon →	Permutation →	Paired t-test
↓	↓	↓	↓
Ignores magnitude of win	fixed Score as per rank of the win	Use exact magnitude of the win	Assumes normal distribution for magnitude of wins
• binomial distribution to Calculate p-value	• normal distribution to calculate p-value	• No distributions/ assumptions	• t distribution to calculate p-value

→ Increasing Sensitivity

Noisy observation → Clean observation

Measures for evaluating binary classifier

- Accuracy : overall how predictions are correct

  Pro : • single number summary
  • fast computation

  Con : • no difference in types of errors
  • Class imbalance
  • model confidence & calibration
  • Threshold training

- Precision: I am confident at identifying positive class

  Pro : • Fast computation
  • Problems where FPs should be minimized

  Con: • Ignore FNs
  • Can be gamed by very rarely predicting the positive class

- Recall: I don't miss many positive instances (Sensitivity)

  Pro: • Fast computation
  • Problems where FNs should be minimized

Page No
Date / /

Con: Ignore FPs

Can be gamed by labelling everything as positive class

- Specificity: I don't miss any negative instances

Pros: • Problems where FPs should be minimised
• Fast computations

Cons: Ignore FNs

• Can be gamed by labelling everything as negative

- Negative Predictive Value: I am confident at identifying negative instances

Pros: Fast computation

Problems where false negatives should be minimised

Con: Ignore FNs

Can be gamed by very rarely predicting the negative class

- F1 Score: I can detect most of predictions instances with high confidence in my predictions

Pros: • Balance btn precision & recall
• Robust to class imbalance

Cons: • Sensitive to low precision or low recall
• Ignores TNs

J =

- Matthews Correlation Coefficient

Pros: Considers all entries in the confusion matrix

Cons: Doesn't differentiate between types of errors

Easy interpretability of other measures

- Why Lloyd's algo inefficient?

• distance calculations redundant

(no need to mention exact distances to say whether this point belongs to this center

Page No:
Date: / /

Given lecture Notes

Properties satisfied by accelerated k-means also
(a) Able to start with any initial centers
(b) Able to use any blackbox distance metric
(c) given the same initial centers, it should always produce the same final centers as the standard algorithm

The only blackbox property that all distance metric d possess is the Δ inequality

↑ gives only upper bounds

To get lower bounds,
- If $d(b,c) \geq 2d(x,b) \Rightarrow d(x,c) \geq d(x,b)$
- $d(x,c) \geq \max(0, d(x,b) - d(x,c))$

l' ← lower bound in prev iteration
$d(x,y) \geq l'$
l ← for current iteration:

$$d(x,b) \geq \max(0, d(x,b') - d(b,b'))$$
$$\geq \max(0, l' - d(b,b'))$$
$$\geq l$$

Elkan's → (i) At start of each iteration, u & l are tight for most points x & centers c
⇒ reducing no. of distance calculations in that iteration

↓

limitation: (i) lower bounds are updated for all points x & centers c ← for each iteration

↓↓

TC : $O(nkc)$

Hamerly algo → (i) 2 distance bounds per data point for its two closest centers

↙ upper bound on distance to the closest center

↘ lower bound on the distance closest to second-closest center